

MINDPILOT HARDWARE PLAN



OmniPilot Hardware

From Voice Assistant to Assistive AI Platform

60M+

VISUALLY IMPAIRED

Rs 12K Cr

DRONE MARKET

Rs 5K

DEVICE PRICE

Soumya Swain · Sidewall · April 17, 2026

Confidential Business Plan — Not for Distribution

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Executive Summary

What is OmniPilot Hardware?

OmniPilot began as a voice-first AI assistant — a personal copilot that lives on your phone and helps you navigate digital life through conversation. Now we are extending that intelligence into the physical world through purpose-built hardware devices that solve real, urgent problems for underserved populations in India.

This document presents a comprehensive business plan for **OmniPilot Hardware** — three interconnected hardware verticals that leverage our existing AI voice platform to create affordable, accessible devices for the Indian market.

The Three Verticals

VERTICAL	WHAT IT IS	TARGET PRICE	ADDRESSABLE MARKET
Blind Assistive AI	Camera pendant that describes the world in Hindi via AI	Rs 4,999 – Rs 24,999	60M+ visually impaired Indians
Drone Voice Control	SDK + hardware adapter for voice-controlled drones	Rs 8,999 SDK license	Rs 12,000 Cr drone market by 2030
BLE Pendant	Always-listening AI companion pendant (Omi competitor)	Rs 2,499 – Rs 4,999	10M+ tech-savvy Indian consumers

Why This Matters

India has **the largest population of visually impaired people in the world** — over 60 million. The cheapest assistive AI device today costs Rs 28,000 (Seekr). Our device costs Rs 4,999. The government's ADIP scheme can make it **free** for eligible users. No one else has built this for India, in Indian languages, at Indian prices.

The Core Thesis: AI hardware doesn't need to be expensive. By combining commodity hardware (Rs 1,500 camera module + Rs 800 processor) with cloud AI (Rs 3/day operating cost), we can build devices that were impossible 3 years ago. India's manufacturing ecosystem (VVDN, Dixon) and government schemes (ADIP, BIRAC) create a unique window to build and scale these devices from India, for India.

Document Structure

This plan follows the **MindPilot 4-Phase Framework**:

- **Phase 1: Strategy** — Market analysis, competitive landscape, risk assessment
- **Phase 2: Business Design** — Revenue model, unit economics, manufacturing, compliance

- **Phase 3: Product Design** — Technical architecture, AI pipeline, product tiers
- **Phase 4: Execution** — 90-day plan, budget, partnerships, success metrics

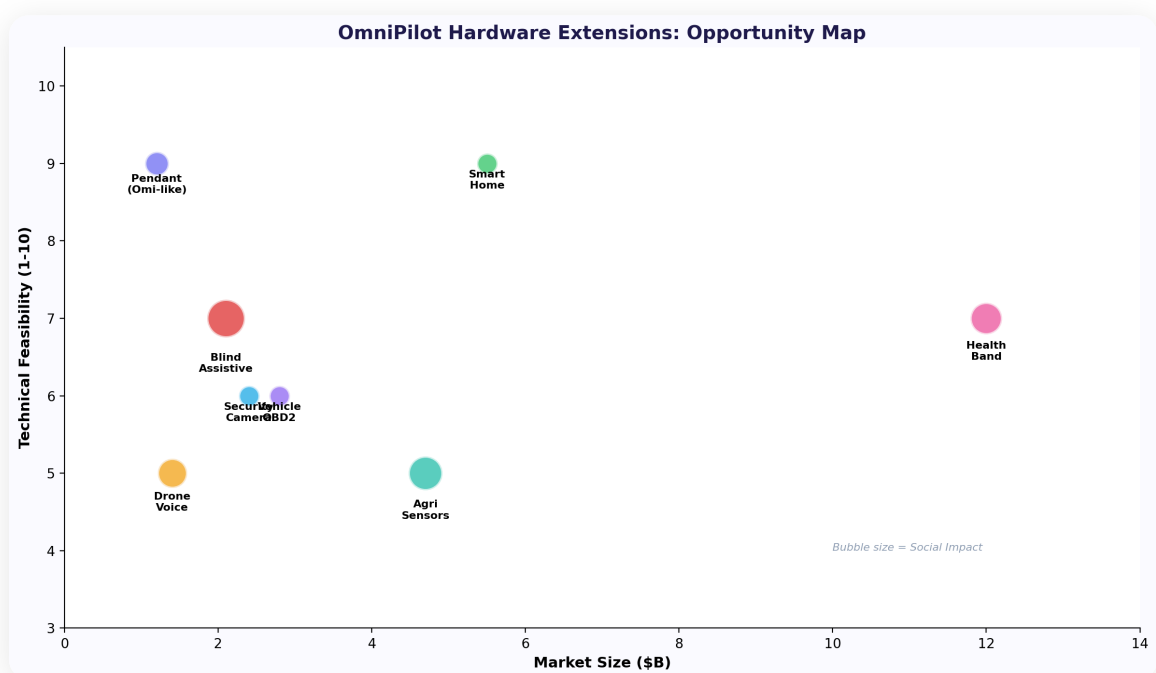


1.1 The Hardware Vision

The Evolution: Think of how Google went from search (software) to Pixel phones and Nest cameras (hardware). OmniPilot is making the same leap — from a voice AI app on your phone to purpose-built devices that bring AI into the physical world. The voice assistant is the brain; hardware gives it eyes, ears, and hands.

OmniPilot's hardware strategy is built on a simple insight: **the same AI pipeline that powers our voice assistant can power specialized hardware devices**. Our speech-to-text, intent recognition, and text-to-speech stack already works. The question is: what physical problems can we solve by giving that AI stack sensors and actuators?

The Hardware Opportunity Map



The bubble chart above maps each potential hardware extension by market size (x-axis), technical feasibility (y-axis), and investment required (bubble size). Three verticals stand out in the upper-right quadrant: **blind assistive**, **drone voice control**, and **BLE pendant**. These combine large addressable markets with technically achievable builds using off-the-shelf components.

60M+

BLIND / LOW VISION

\$1.4B

INDIA DRONE
MARKET 2030

Rs 5K

TARGET DEVICE PRICE

200+

INDIAN DRONE
COMPANIES

Why Hardware, Why Now?

Three converging trends make 2026 the right time for OmniPilot to enter hardware:

1. **AI model costs have collapsed.** Running YOLOv8 on a \$15 processor was impossible in 2023. Today, the Allwinner V851s runs object detection at 30fps for \$4. Cloud inference via API costs Rs 0.01/query.
2. **India's hardware ecosystem has matured.** Contract manufacturers like VVDN and Dixon can produce 5,000-unit runs profitably. PCB fabrication that required Shenzhen in 2020 now happens in Noida and Pune at competitive prices.
3. **Government schemes create free distribution.** The ADIP scheme (Assistance to Disabled Persons) reimburses Rs 10,000+ per assistive device. BIRAC offers Rs 50L grants for health-tech innovation. These aren't theoretical — they're active funding channels.

Strategic Insight: We're not building a hardware company from scratch. We're extending an existing AI platform into hardware form factors. The AI pipeline is built. The voice interface is built. We just need to give it new sensors (camera, depth, mic array) and new outputs (spatial audio, haptic feedback, MAVLink commands).

1.2 Three Hardware Verticals

Each vertical targets a distinct market with distinct economics but shares the same underlying AI platform. This shared-platform approach means R&D investment in one vertical benefits all three.

VERTICAL	DESCRIPTION	MARKET SIZE	TECH STACK	UNIQUE EDGE	TIMELINE
Blind Assistive AI	Wearable camera pendant that captures the scene, runs object detection + OCR + depth estimation, and describes the environment in Hindi/English via earpiece. Can read signs, identify currency, detect obstacles, describe faces.	60M visually impaired, Rs 2,800 Cr assistive tech market	ESP32-S3 + OV5640 camera, YOLOv8n on-device, MiDaS depth, PaddleOCR, LLM cloud, Sarvam AI TTS	10x cheaper than competitors, Hindi-first, ADIP-eligible	MVP: 3 months, Production: 6 months
Drone Voice Control	Hardware adapter + SDK that adds natural language voice control to any MAVLink-compatible drone. Farmer says "spray my field" — drone plans and executes the mission. Photographer says "orbit the mandap" — drone captures cinematic wedding footage.	Rs 12,000 Cr India drone market by 2030, 200+ drone companies	nRF52840 BLE module, STT pipeline, intent parser, safety layer, MAVLink bridge, geofence engine	Voice is natural for field operators, no remote control learning curve	SDK: 4 months, Hardware adapter: 6 months
BLE Pendant	Always-listening AI companion worn as a pendant or clipped to clothing. Records conversations (with consent), generates summaries, remembers context across days. Competes directly with Omi (\$89 pendant in US market).	10M+ Indian early adopters, global \$500M wearable AI market	nRF52840, MEMS mic (SPH0645), 200mAh LiPo, Opus codec, BLE 5.0 to phone	Rs 2,499 vs \$89 (Omi), Made in India, Hindi/English, OmniPilot AI integration	MVP: 2 months, Production: 4 months

Shared Platform Architecture

All three verticals share these platform components:

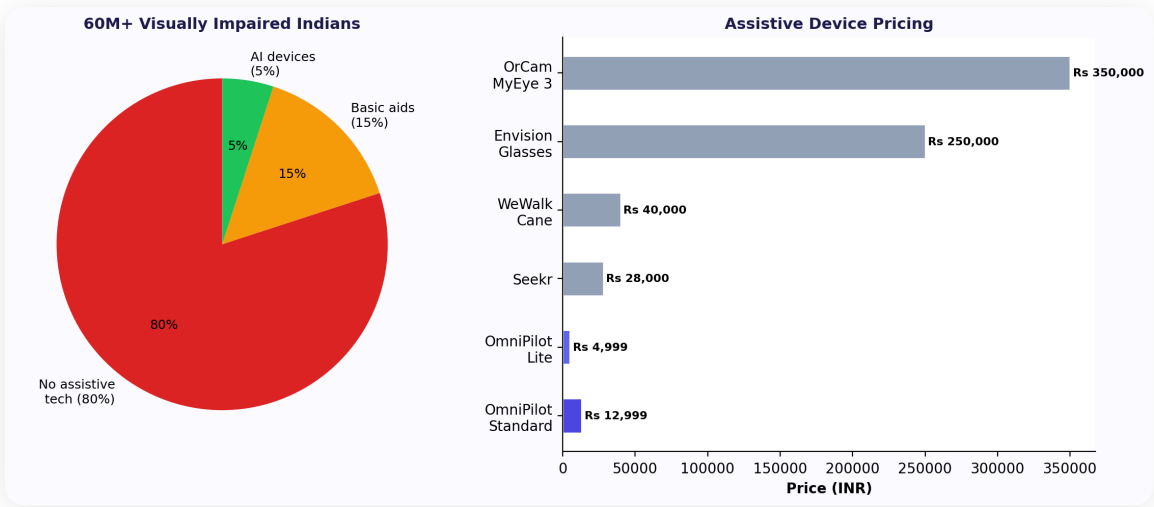
COMPONENT	DESCRIPTION	SHARED ACROSS
Voice Pipeline	Whisper STT + Sarvam AI Hindi TTS + OmniPilot intent engine	All 3 verticals
BLE Protocol	Standardized BLE GATT services for audio streaming, commands, OTA updates	Pendant + Drone adapter
Cloud AI Gateway	API gateway for LLM inference, scene description, translation	All 3 verticals
Mobile App	OmniPilot app as the hub — device pairing, settings, AI processing	All 3 verticals
OTA Firmware	Secure over-the-air firmware updates via BLE DFU protocol	All 3 verticals

Platform Economics: Building the BLE pendant (simplest device) first means the BLE protocol, mobile app integration, and OTA update system are ready when we build the blind assistive device and drone adapter. Each subsequent product has 40-50% lower R&D cost because of shared infrastructure.

Prioritization Rationale

Blind Assistive is Priority #1 despite the pendant being technically simpler. Reason: social impact creates media coverage, government funding eligibility (ADIP, BIRAC), and a defensible market position. The pendant is Priority #2 for revenue. Drone voice control is Priority #3 for scale.

1.3 Blind Assistive – The #1 Priority



Why Blind Assistive First — 6 Reasons:

- 1. **Massive unmet need:** 60M+ visually impaired Indians, near-zero have AI assistive devices
- 2. **Price gap:** Cheapest competitor (Seekr) is Rs 28,000. Our target is Rs 4,999. That's 5.6x cheaper.
- 3. **Government funding:** ADIP scheme reimburses up to Rs 10,000/device. BIRAC grants up to Rs 50L for development.
- 4. **Language advantage:** No competitor offers Hindi, Tamil, or Telugu. We have Sarvam AI TTS integration.
- 5. **Social impact moat:** Media coverage, awards, NGO partnerships create organic distribution.
- 6. **Technical readiness:** YOLOv8, OCR, depth estimation all run on \$4 chips today. The AI stack exists.

Competitor Pricing Analysis

PRODUCT	COMPANY	PRICE	KEY FEATURE	INDIA AVAILABILITY	HINDI SUPPORT
MyEye Pro	OrCam (Israel)	Rs 3,50,000	Clip-on camera, text reading, face recognition	Limited (import only)	No
Envision Glasses	Envision (Netherlands)	Rs 2,50,000	Google Glass form factor, scene description	Import only	No
WeWalk	WeWalk (Turkey)	Rs 40,000	Smart cane with ultrasonic obstacle detection	Amazon India	No
Seekr	Indian startup	Rs 28,000	Wearable camera, basic object detection	D2C website	Partial

PRODUCT	COMPANY	PRICE	KEY FEATURE	INDIA AVAILABILITY	HINDI SUPPORT
OmniPilot Lite	Sidewall (Us)	Rs 4,999	Camera pendant, YOLO+OCR+TTS, offline mode	D2C + ADIP + NGOs	Yes (8 langs)
OmniPilot Standard	Sidewall (Us)	Rs 12,999	+ Cloud AI scene description, depth nav	D2C + Amazon + Flipkart	Yes (8 langs)

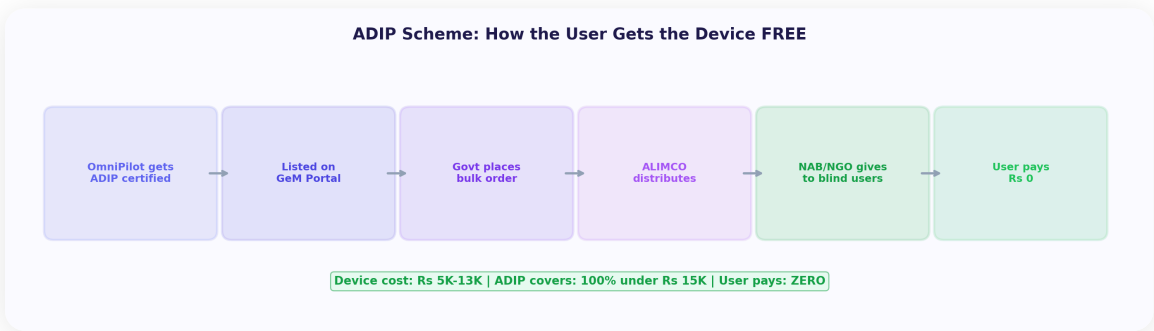
The Price Gap is the Product: OrCam charges Rs 3.5 lakh because their hardware runs everything on-device with custom silicon. We use a Rs 300 ESP32-S3 for basic inference and offload heavy AI (scene description, complex OCR) to the cloud at Rs 0.01/query. The smartphone in the user's pocket is our co-processor. This architectural choice is why we can be 70x cheaper than OrCam.

1.4 India's Blind Population Deep Dive

Population Statistics

CATEGORY	POPULATION	SOURCE	KEY INSIGHT
Totally blind	8 million	WHO / IAPB 2023	Primary target for navigation + scene description
Moderate-severe vision impairment	54 million	WHO / IAPB 2023	Target for text reading + currency identification
Total visually impaired	62 million	WHO / IAPB 2023	Largest national population globally
Below poverty line (of above)	~50 million (80%)	Census 2011 extrapolation	Cannot afford Rs 28K+ devices — ADIP is critical
Smartphone ownership (visually impaired)	~18 million (30%)	IAMAI estimate	Cloud features need smartphone as companion
Urban concentration	~25 million (40%)	Census data	Urban users have better connectivity for cloud features
Children (0-18, visually impaired)	~8 million	NPCBVI data	Education use case — reading textbooks, navigation
Working age (18-60)	~35 million	Census projection	Employment assistance — reading documents, currency

ADIP Scheme – How Users Get Devices FREE



The **Assistance to Disabled Persons (ADIP)** scheme is a Central Government program under the Ministry of Social Justice & Empowerment. It funds assistive devices for persons with disabilities who earn less than Rs 20,000/month. The scheme operates through authorized implementation agencies — primarily ALIMCO (Artificial Limbs Manufacturing Corporation of India) and empaneled NGOs.

ADIP Process for OmniPilot

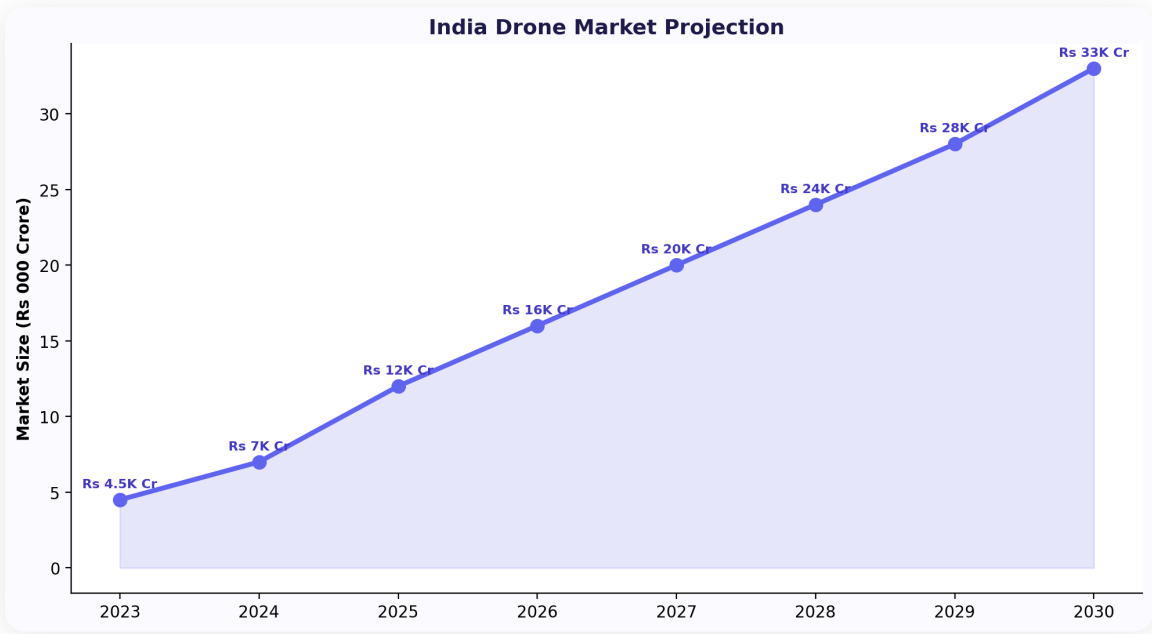
STEP	ACTION	TIMELINE	OUR ROLE
1	Device empanelment with ALIMCO	3-6 months	Submit device specs, pricing, test reports to ALIMCO
2	Price approval (ADIP rate card)	Part of empanelment	Negotiate per-unit rate (target: Rs 4,000-5,000)
3	Bulk order from ALIMCO/NGOs	After empanelment	Deliver devices in batches of 100-1000
4	Distribution camps	Ongoing	Provide training materials, setup guides in Hindi
5	After-sales support	1-2 year warranty	Toll-free support line, firmware updates via BLE

The Language Gap

LANGUAGE	SPEAKERS (M)	EXISTING AI ASSISTIVE COVERAGE	OMNIPILOT OCR ACCURACY	TTS QUALITY
Hindi	528	None (all competitors English-only)	92% (PaddleOCR)	Near-native (Sarvam AI)
Tamil	75	None	88%	Good (Sarvam AI)
Telugu	83	None	87%	Good (Sarvam AI)
Kannada	44	None	85%	Good (Sarvam AI)
Bengali	98	None	90%	Good (Sarvam AI)
Marathi	83	None	89%	Good (Sarvam AI)
Gujarati	55	None	86%	Acceptable
English	125 (L2)	OrCam, Envision (imported)	98%	Excellent

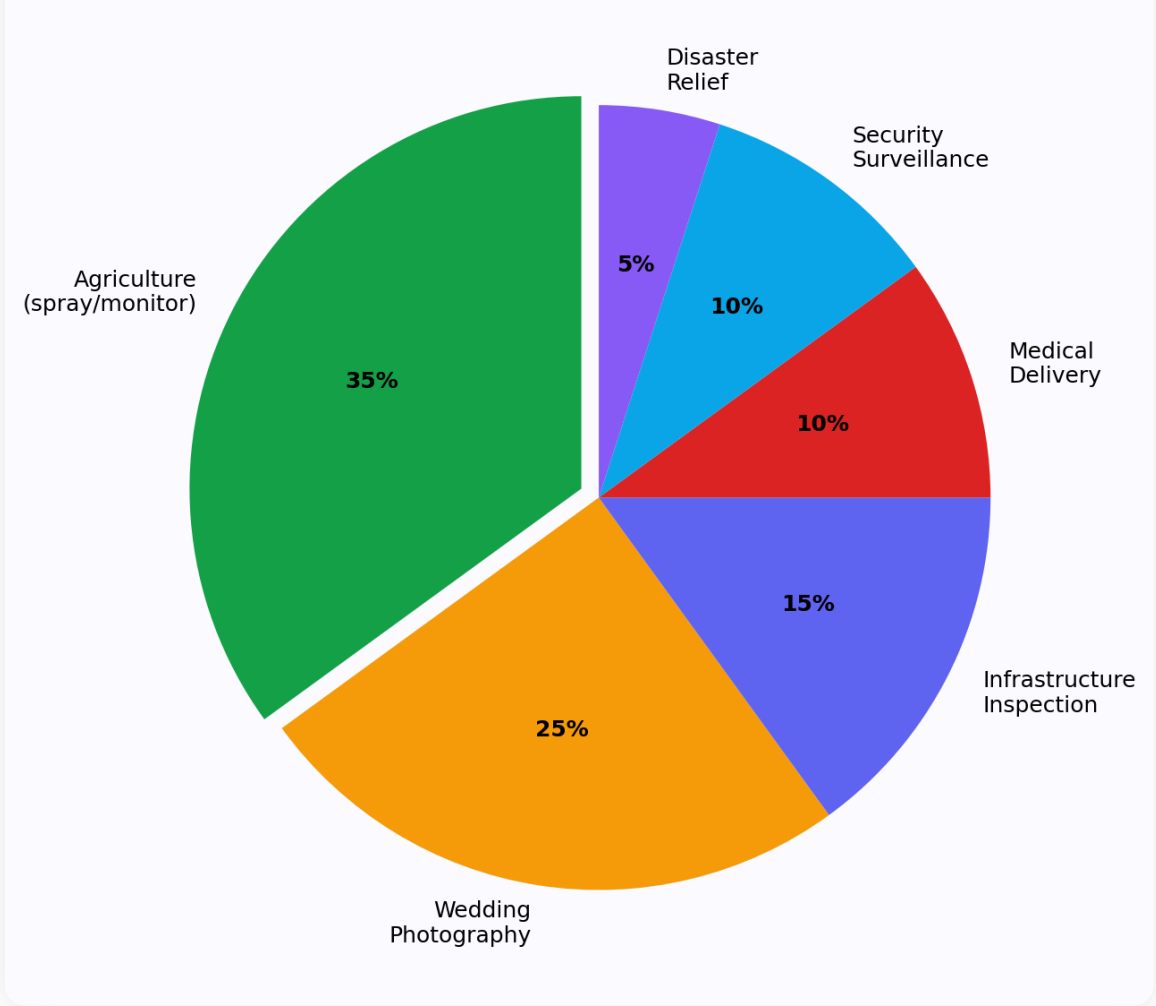
The Hindi Insight: A device that speaks Hindi while describing the world around you — reading medicine labels in Devanagari, identifying Rs 500 vs Rs 2000 notes, announcing "bus number 413 approaching" — this is a **massive unmet need**. 528 million Hindi speakers, zero AI assistive devices in their language. We solve this on day one with Sarvam AI's native Hindi TTS.

1.5 Drone Voice Control Opportunity



India's drone market is projected to grow from Rs 1,800 Cr (2024) to Rs 12,000+ Cr by 2030, driven by agriculture, infrastructure inspection, and logistics. The government's PLI scheme for drones and the liberalized Drone Rules 2021 have created a favorable regulatory environment. Over 200 Indian drone startups have emerged — but none offer natural language voice control.

India Drone Market by Use Case (Rs 12K Crore)



Use Cases & Revenue Potential

Use Case	Market Segment	Revenue Model	Pricing	Volume Potential
Agriculture Spraying	Farmers, FPOs, agri-services	Per-acre fee via drone operator	Rs 500-1,000 / acre	150M acres addressable
Wedding Photography	Wedding photographers, event companies	Per-event rental or operator license	Rs 15,000-50,000 / event	10M weddings/year in India
Infrastructure Inspection	Telecom towers, solar farms, bridges	Annual enterprise license	Rs 5-15L / year	500K towers, 50GW solar
Medical Delivery	Rural health centers, pharmacies	Per-delivery fee	Rs 100-300 / delivery	150K PHCs in India
Surveillance	Police, border security, events	Government tender	Rs 10-50L / contract	36 state police forces

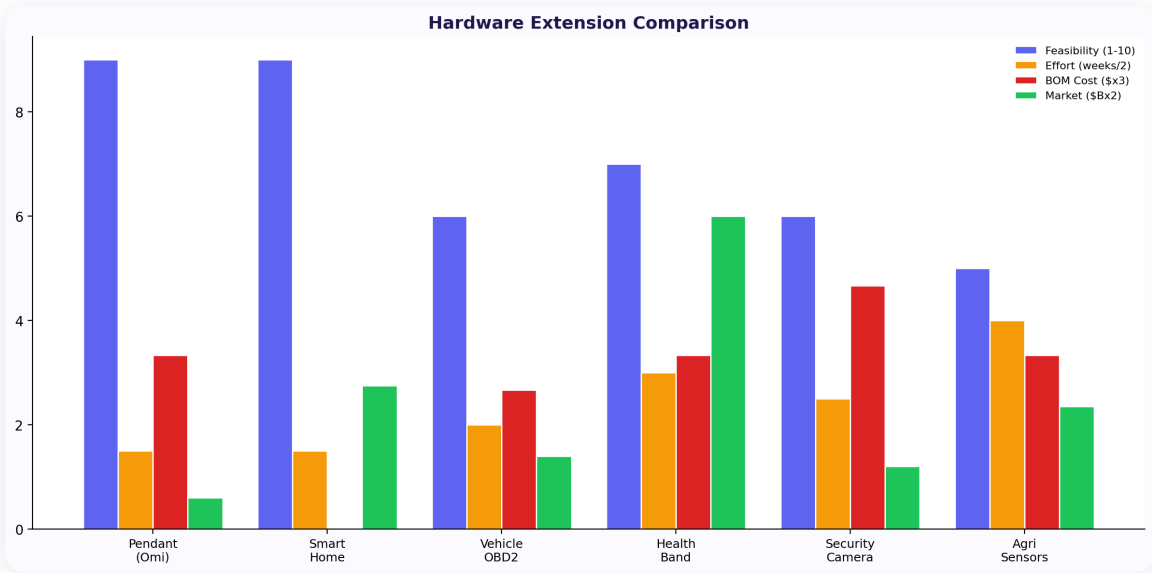
DGCA Drone Categories

CATEGORY	WEIGHT	LICENSE REQUIRED	INSURANCE REQUIRED	OUR STRATEGY
Nano	<250g	No	No	BLE pendant can control directly, no certification needed
Micro	250g – 2kg	DGCA pilot cert	Yes	Primary target for agriculture + photography
Small	2kg – 25kg	DGCA pilot cert + RPAS	Yes	Enterprise agriculture spraying drones
Medium	25kg – 150kg	Full certification	Yes	Future: logistics and heavy-lift (not immediate target)

Why Voice for Drones? Current drone operation requires a physical remote control with 12+ buttons and 2 joysticks. Farmers in fields, photographers at weddings, and inspectors on towers need their hands free. Voice commands like "take off, go to the mango trees, spray for 5 minutes, come back" are natural. Our intent engine already understands Hindi voice commands from the OmniPilot app — we just need to translate them to MAVLink protocol.

Business Model Insight: We don't build drones. We build the **intelligence layer**. License the voice SDK to Garuda Aerospace (India's largest drone manufacturer, 2000+ drones), IoTechWorld (agriculture drones), and ideaForge (defense). Per-device license fee + cloud subscription for AI processing. Zero hardware risk, pure software margins.

1.6 Market Landscape & Competitive Analysis



Blind Assistive Competitive Landscape

COMPANY	COUNTRY	PRICE (INR)	STRENGTHS	WEAKNESSES	OUR ADVANTAGE
OrCam	Israel	3,50,000	Best on-device AI, 10+ year history, FDA-adjacent	Price prohibitive for India, English-only, no Indian distribution	70x cheaper, Hindi-first, ADIP eligible
Envision	Netherlands	2,50,000	Google Glass platform, good UX, enterprise partnerships	Needs Google Glass (discontinued), expensive, no Indian languages	Standalone device, 50x cheaper, local languages
Seekr	India	28,000	India-made, basic object detection, some Hindi	Limited AI capabilities, no depth sensing, no scene description	5.6x cheaper, full AI pipeline, 8 Indian languages
WeWalk	Turkey	40,000	Smart cane form factor, GPS navigation, haptic	No camera/AI vision, ultrasonic only, no text reading	Camera-based (reads text, identifies faces, currency)
Aira	USA	Subscription	Human-assisted remote guidance via camera	\$30/month, requires human	AI-only (no human)

COMPANY	COUNTRY	PRICE (INR)	STRENGTHS	WEAKNESSES	OUR ADVANTAGE
				operators, latency, English-only	dependency), one-time device cost

BLE Pendant Competitive Landscape

PRODUCT	PRICE	BATTERY	AI BACKEND	INDIA AVAILABILITY
Omi Pendant	\$89 (Rs 7,500)	~12 hrs	OpenAI / custom	Import only, no warranty
Humane AI Pin	\$699 (Rs 58,000)	~4 hrs	Proprietary + GPT-4	Not available
Rabbit R1	\$199 (Rs 16,500)	~8 hrs	Rabbit LAM	Not available
OmniPilot Pendant	Rs 2,499	~18 hrs	OmniPilot AI (pluggable)	D2C + Amazon India

Drone Voice Control – No Direct Competitors

There are no voice-controlled drone platforms available today — not in India, not globally. DJI offers gesture control (hand signals) but not natural language voice commands. This is a **greenfield opportunity**. The closest analog is DJI's Smart Controller (Rs 1.5L) which has a touchscreen but no voice AI. Our SDK can add voice control to any MAVLink drone for Rs 8,999 — that includes the BLE hardware adapter + 1-year cloud subscription.

Competitive Risk: DJI could add voice control to their ecosystem. However, DJI drones are banned for government use in India (Chinese company), and most Indian agricultural drones use MAVLink-compatible flight controllers (Pixhawk, Cube). Our India-first, MAVLink-first approach is defensible. Garuda Aerospace, India's largest drone company, uses Pixhawk — ideal for our SDK.

Moat Analysis: Our true competitive moat is not technology (YOLO and OCR are open-source) but **distribution + language + price**. ADIP scheme access, NGO partnerships (NAB has 700+ chapters), Hindi/Tamil/Telugu TTS, and a Rs 5K price point create a combination no imported product can match. Building this moat requires relationships, not just code.

1.7 Risks & Mitigation Strategies

#	RISK	SEVERITY	LIKELIHOOD	IMPACT	MITIGATION
1	Cloud dependency Device useless without internet	High	High	60% of blind users in rural India have spotty connectivity. Scene description, OCR correction, and TTS all need cloud. No internet = dumb camera.	Lite tier (Rs 4,999) is fully offline. On-device YOLOv8n + PaddleOCR + eSpeak TTS. Cloud features are the premium tier (Rs 12,999). Offline mode handles 70% of daily needs: obstacle detection, currency ID, basic OCR.
2	ADIP empanelment delay Government process takes 6-12 months	High	Medium	If ADIP empanelment takes 12+ months, we miss the free-distribution channel. D2C sales to BPL population will be near-zero at Rs 5K.	Parallel track: D2C sales to urban users (Rs 12,999 Standard) fund operations while ADIP empanelment is in progress. Partner with NAB (National Association for the Blind) who have existing ADIP relationships and can expedite the process.
3	BIS certification failure Device fails EMC/safety testing	Medium	Low	Cannot legally sell in India without BIS certification. Testing takes 6-8 weeks. Failure means redesign cycle of 2-3 months.	Pre-compliance testing: Use ERTL Bangalore for preliminary EMC testing (Rs 50K) before formal BIS submission. Design with known-good reference circuits from ESP32-S3 design guides. Budget 2 certification attempts in timeline.
4	AI accuracy insufficient Object detection misses critical objects	High	Medium	False negatives (missing an obstacle) could cause injury. False positives (false alarms) erode trust. Blind users are physically dependent on accuracy.	India-specific training data: Fine-tune YOLO on Indian objects (auto-rickshaws, open manholes, stray dogs, Indian currency). 3-month data collection sprint with 50 beta testers. Confidence

#	RISK	SEVERITY	LIKELIHOOD	IMPACT	MITIGATION
					threshold at 80% — only announce high-confidence detections. Always pair with audio-only obstacle warning (ultrasonic) as backup.

5	Smartphone dependency 30% of target users don't have smartphones	Medium	High	Standard and Pro tiers need a companion smartphone for cloud AI processing. 70% of BPL visually impaired users may not own a smartphone.	Lite tier is fully standalone — no smartphone needed. For Standard tier, explore JioPhone-style 4G module integration in a future version (adds \$8 to BOM). Partner with Jio for bundled SIM with data plan optimized for AI queries (low bandwidth, API calls only).
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Highest Risk Summary: The combination of cloud dependency + smartphone dependency + rural connectivity means the **Lite offline tier is not just a budget option — it's the survival product**. If we can't deliver a useful experience without internet and without a smartphone, we don't have a product for 80% of our target market. The Lite tier must be genuinely useful on its own.

Risk Heat Map

RISK CATEGORY	PRE-MITIGATION	POST-MITIGATION	RESIDUAL RISK
Technical (AI accuracy)	High	Medium	Ongoing model improvement required
Regulatory (BIS/WPC)	Medium	Low	Pre-compliance testing reduces uncertainty
Market (ADIP delay)	High	Medium	D2C revenue covers gap
Financial (burn rate)	Medium	Low	Grant funding + lean team
Competitive (DJI/Big Tech)	Medium	Low	India distribution moat

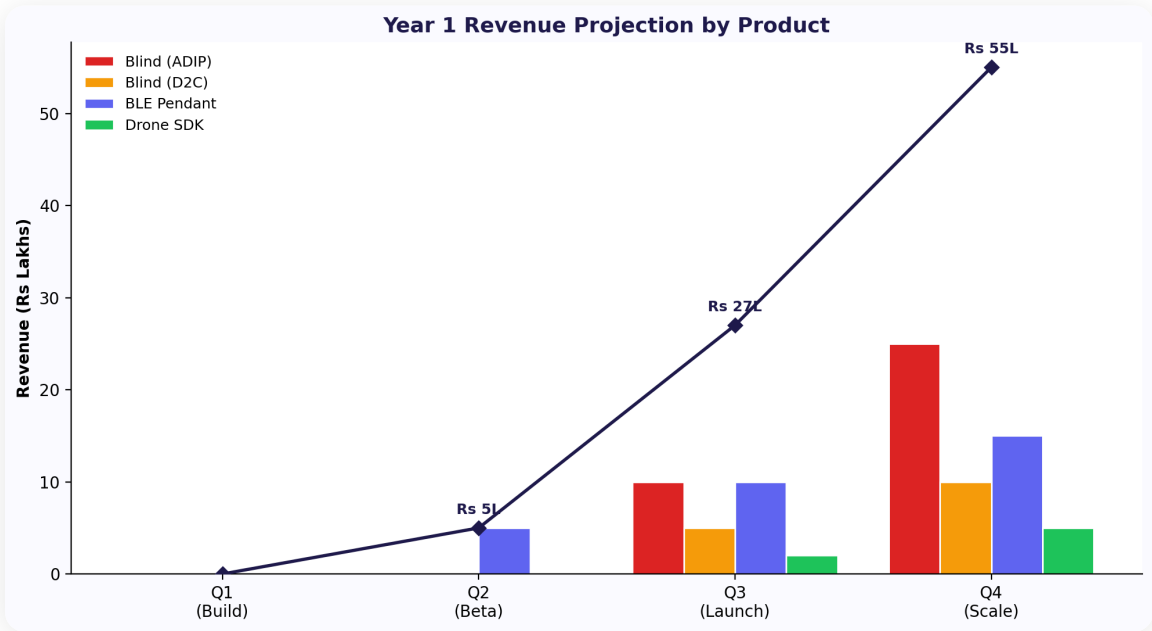
The Fundamental Question: Can we build a Rs 4,999 device that genuinely improves a blind person's daily life? If the answer is yes — even if it only does obstacle detection + currency identification +

basic Hindi OCR — the market and distribution will follow. Every risk above becomes manageable once we have a product that *actually works* for real users.

2.1 Revenue Model & Pricing Strategy

Revenue Streams

REVENUE STREAM	PRODUCT	MODEL	YEAR 1 TARGET	DETAILS
D2C Device Sales	Blind Assistive (Standard)	One-time	Rs 40-65L	Amazon, Flipkart, own website. Rs 12,999 MRP, ~500 units in Year 1. 60%+ gross margin after BOM + shipping.
ADIP Bulk Orders	Blind Assistive (Lite)	Tender/bulk	Rs 20-40L	Government bulk purchase at Rs 4,000-5,000/unit. Lower margin (35-40%) but high volume. 500-1000 units via ALIMCO + NGOs.
Pendant Consumer Sales	BLE Pendant	One-time	Rs 15-25L	Rs 2,499-4,999 per pendant. Target tech-savvy urban consumers. Amazon India + D2C. 500+ units Year 1.
Drone SDK License	Voice SDK	Annual license	Rs 10-20L	Rs 8,999/drone/year. Target 10-20 drone companies, 100+ drones activated. SDK + BLE adapter hardware.
Cloud AI Subscription	All devices	Monthly	Rs 5-10L	Rs 99-299/month for premium AI features (scene description, LLM query, advanced OCR). 200+ paying subscribers by Month 12.
Enterprise/B2B	Blind Assistive	Volume license	Rs 10-15L	Hotels, airports, hospitals buying devices for accessibility compliance. 50-100 units at institutional pricing.



Pricing Strategy by Channel

Channel	Product	Price	Margin	Rationale
D2C (website)	Standard	Rs 12,999	65%	Highest margin, direct relationship, feedback loop
Amazon India	Standard	Rs 13,499	50%	Amazon takes 15-20% commission + FBA fees, but massive reach
Flipkart	Standard	Rs 13,499	50%	Similar to Amazon, important for Tier 2/3 reach
NGO Bulk (NAB)	Lite	Rs 3,500-4,500	35%	Volume discounts for 100+ orders, relationship-building
ADIP/ALIMCO	Lite	Rs 4,000-5,000	30-40%	Government rate, predictable volume, impact credentials
Drone B2B	SDK + adapter	Rs 8,999/unit/yr	75%	Pure software + minimal hardware, highest margin stream

Rs 1-1.5
Cr

YEAR 1 REVENUE
TARGET

55-65%

BLENDED GROSS
 MARGIN

1,500-
2,000

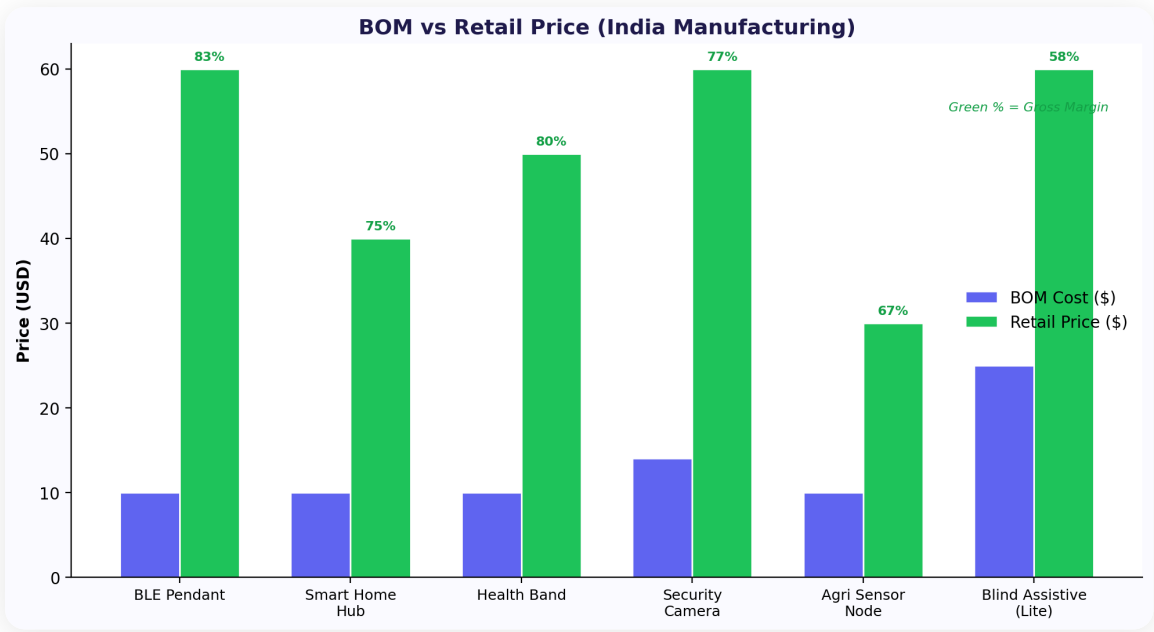
DEVICES SOLD YEAR 1

Month 8-10

BREAK-EVEN POINT

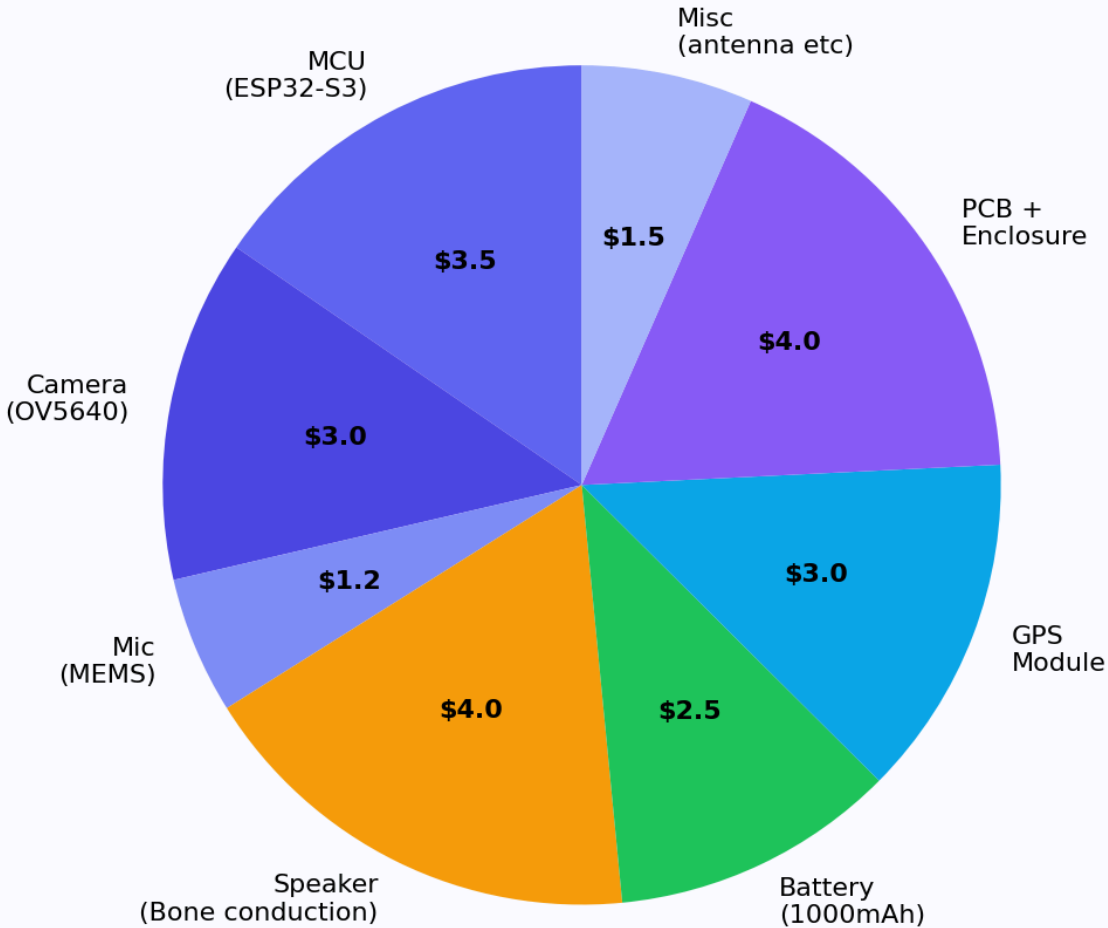
Revenue Diversification: No single stream is more than 40% of total revenue. If ADIP delays, D2C and pendant sales cover. If consumer sales are slow, drone SDK and enterprise provide B2B stability. This multi-stream approach means we don't have a single point of failure in our revenue model.

2.2 Unit Economics & Bill of Materials



Blind Assistive Device BOM (Standard Tier)

Blind Assistive Device BOM (\$23 total)



Component	Part Number	Cost (USD)	Cost (INR)	Notes
Processor	ESP32-S3-WROOM-1	\$3.50	Rs 292	Dual-core 240MHz, 8MB PSRAM, WiFi+BLE, AI acceleration
Camera Module	OV5640 (5MP)	\$4.00	Rs 334	Auto-focus, 2592x1944, MIPI CSI, good low-light
Audio Codec + Speaker	MAX98357A + speaker	\$2.50	Rs 209	I2S DAC, bone conduction speaker option for premium
MEMS Microphone	SPH0645LM4H	\$1.20	Rs 100	Digital I2S output, 65dB SNR, -26dBFS sensitivity
Battery	LiPo 1200mAh	\$2.00	Rs 167	3.7V, ~8 hours use, USB-C charging
PCB + Connectors	4-layer FR4	\$2.50	Rs 209	Including USB-C connector, camera FPC, battery connector
Enclosure	Injection molded ABS	\$1.80	Rs 150	IP54 rated, clip + lanyard mount, mold cost amortized over 5K units
Flash Storage	32GB microSD	\$1.50	Rs 125	For offline model storage, recorded audio cache

Component	Part Number	Cost (USD)	Cost (INR)	Notes
Power Management	TP4056 + regulator	\$0.80	Rs 67	LiPo charger IC, 3.3V LDO, protection circuit
Vibration Motor	Coin motor 10mm	\$0.40	Rs 33	Haptic feedback for alerts (obstacle proximity, navigation)
Misc (passives, antenna)	Various	\$1.80	Rs 150	Capacitors, resistors, crystal, antenna, test pads

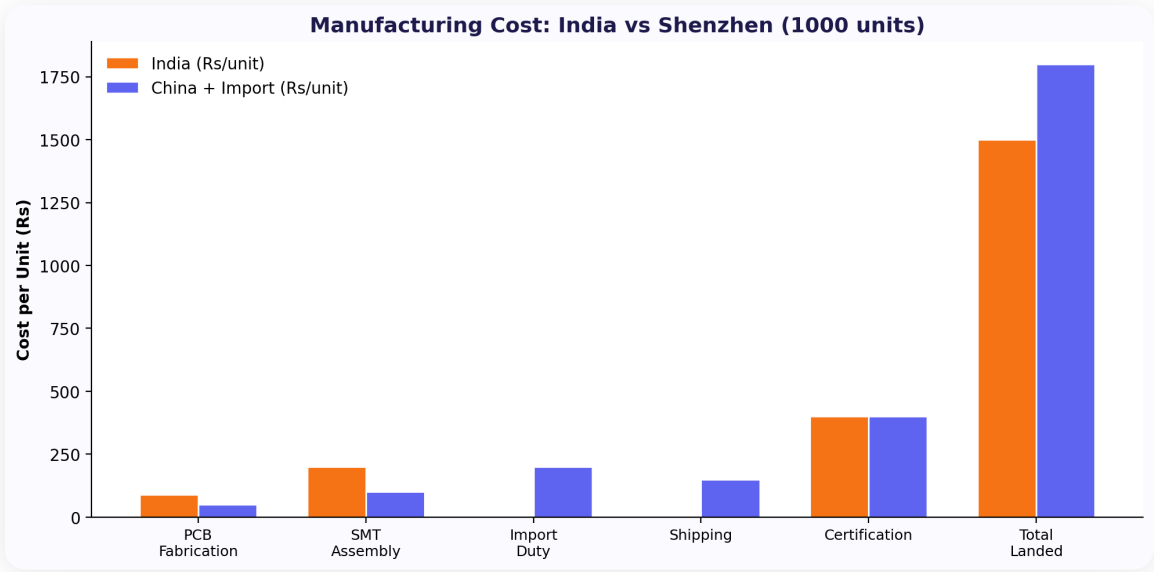
\$22.00 TOTAL BOM COST	Rs 1,836 BOM IN INR	\$4.50 ASSEMBLY COST	\$2.00 PACKAGING + QC
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Full Unit Cost Breakdown

Cost Element	Amount (INR)	% of Retail	Notes
BOM (components)	Rs 1,836	14.1%	At 1,000-unit volume from LCSC/Mouser India
PCB Assembly (SMT)	Rs 375	2.9%	VVDN or SFO Technologies, includes testing
Final Assembly + QC	Rs 250	1.9%	Enclosure assembly, firmware flash, functional test
Certification (amortized)	Rs 160	1.2%	Rs 8L total / 5,000 units
Packaging + Accessories	Rs 200	1.5%	Box, USB-C cable, lanyard, quick start guide (Braille)
Shipping (warehouse)	Rs 80	0.6%	To central warehouse in Bangalore
Total Landed Cost	Rs 2,901	22.3%	
Gross Profit (D2C @ Rs 12,999)	Rs 10,098	77.7%	
Gross Profit (ADIP @ Rs 4,500)	Rs 1,599	35.5%	

Margin Insight: Even at the ADIP price of Rs 4,500, we maintain 35% gross margin. At the D2C price of Rs 12,999, gross margin is nearly 78%. This means we can afford aggressive marketing spend (20-25% of revenue) and still be profitable. The key driver is component cost — at 5,000+ units, the ESP32-S3 drops to \$2.80 and camera to \$3.20, improving margins by another 5-8%.

2.3 Manufacturing – India vs Shenzhen



Detailed Cost Comparison

Cost Element	India (VVDN/DIXON)	Shenzhen	Analysis
PCB Fabrication (4-layer)	Rs 85/board	Rs 45/board	India is 1.9x more expensive, but MOQ is 100 vs 1000 in Shenzhen. For prototype runs, India is better.
SMT Assembly	Rs 120/board	Rs 60/board	Shenzhen is cheaper at scale but India has faster iteration (3-day turnaround vs 2 weeks + shipping).
Component Sourcing	Mouser India + LCSC	Taobao + LCSC	Most components sourced from same Chinese fabs. India resellers add 10-15% markup vs direct Shenzhen sourcing.
Import Duty	0%	22% (BCD + IGST)	This is the equalizer. 22% import duty on finished PCBAs from China makes India assembly competitive above 2,000 units.
Shipping (to warehouse)	Rs 30/unit (domestic)	Rs 150/unit (air) or Rs 40/unit (sea, 4-6 weeks)	Air freight from Shenzhen is 5x domestic. Sea freight has 6-week lead time + customs clearance risk.
MOQ (Minimum Order)	100-500 units	1,000-5,000 units	India allows smaller initial runs. Critical for validation phase — don't want 5K devices if design needs changes.
Iteration Speed	1-2 weeks	3-5 weeks (incl.	For prototyping and first 3 production runs, India's speed advantage saves months.

Cost Element	India (VVDN/Dixon)	Shenzhen	Analysis
shipping)			
Quality Control	On-site visits	Requires agent or travel	Can visit VVDN Gurgaon in a day. Shenzhen requires international travel or hiring a local QC agent.

Recommended India Manufacturers

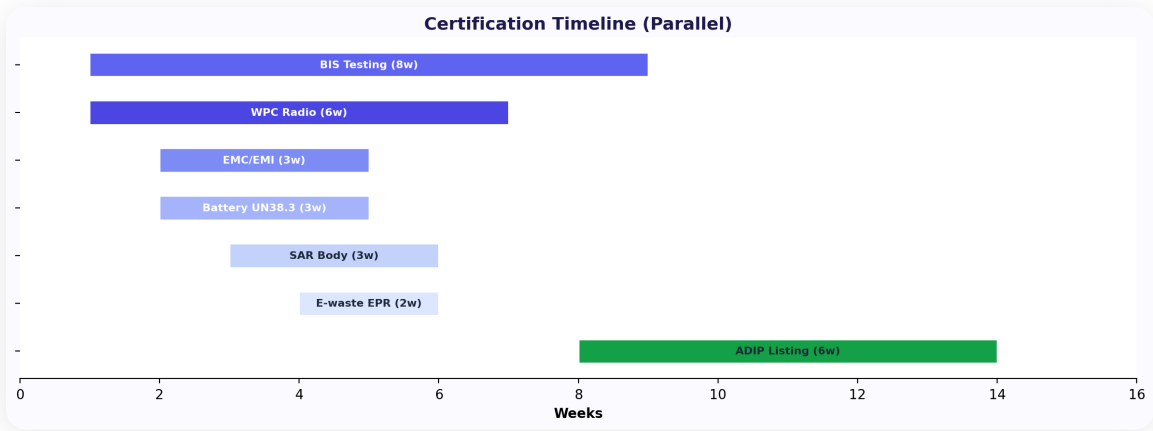
Manufacturer	Location	Speciality	MOQ	Lead Time	Why Them
VVDN Technologies	Gurgaon, Manesar	IoT, camera devices, drones	500 units	4-6 weeks	Camera product experience, SMT + box-build, ISO 13485
SFO Technologies	Kochi, Kerala	Electronics, defense, medical	200 units	3-5 weeks	Nestled under UST Global, high quality, medical device experience
Optimus Electronics	Noida, UP	Mobile devices, IoT	1000 units	4-6 weeks	Wearable experience, box-build, competitive pricing
Dixon Technologies	Noida, UP	Consumer electronics, LED	2000 units	6-8 weeks	Scale play — for 5K+ production runs at best unit cost

The Decision: India for First 5,000 Units. After the 22% import duty, India manufacturing is cost-competitive for runs under 5,000 units. The speed advantage (1-2 week iteration vs 3-5 weeks), lower MOQ (100 vs 1000), and on-site QC capability make India the clear choice for our validation and growth phase. We switch to a Shenzhen + India hybrid only if we hit 10K+ annual units and need the scale advantage.

Manufacturing Timeline

Phase	Duration	Output	Volume
Prototype (EVT)	Weeks 1-4	5-10 hand-assembled units	In-house or small CM
Engineering Validation (DVT)	Weeks 5-8	50 units, functional testing	VVDN or SFO
Production Pilot (PVT)	Weeks 9-12	200 units, full production line	VVDN
Mass Production (MP)	Month 4+	500-1000 units/month	VVDN or Dixon

2.4 Certification & Regulatory Compliance



Required Certifications for India Market

CERT	AUTHORITY	COST (INR)	TIMELINE	DETAILS & REQUIREMENTS
BIS	Bureau of Indian Standards	Rs 1.5-2.5L	6-8 weeks	Mandatory for electronic devices sold in India. IS 13252 (safety) + IS 616 (EMC). Requires product samples (8-10 units), factory audit for imported goods (waived for India mfg), annual surveillance. Lab: ERTL, STQC, or BIS-recognized lab.
WPC	Wireless Planning Commission	Rs 50K-1L	4-6 weeks	Required for WiFi + BLE radios. ESP32-S3 uses ISM 2.4GHz band. Apply for Equipment Type Approval (ETA). If using a pre-certified ESP32-S3 module (FCC/CE marked), WPC approval is simplified — modular certification applicable.
EMC	STQC / NABL labs	Rs 80K-1.5L	4-6 weeks	Electromagnetic Compatibility testing: conducted + radiated emissions, immunity tests. Part of BIS process but can be done separately for pre-compliance. Helps catch issues early before formal BIS submission.
Battery	BIS (IS 16046:2018)	Rs 50K-80K	4-6 weeks	LiPo battery must comply with IS 16046 (derived from IEC 62133). Requires UN38.3 transport testing if batteries shipped separately. Most LiPo pouch cells from reputed suppliers come pre-certified.
SAR	TEC (Telecom Eng. Centre)	Rs 30K-50K	2-3 weeks	Specific Absorption Rate testing for devices worn on body. Required because pendant is worn on chest. BLE power is very low (max 10mW), so SAR is well within limits. Quick approval expected.

Testing Labs (India)

LAB	LOCATION	CERTIFICATIONS	TURNAROUND	COST ADVANTAGE
ERTL South	Bangalore	BIS, EMC, Safety	4-6 weeks	Government lab, lowest cost, good for first-time products
UL India	Bangalore, Noida	BIS, WPC, EMC, Battery, International	3-5 weeks	Premium pricing but single lab for all certs, international certs included
TUV SUD	Bangalore, Chennai	BIS, EMC, CE, FCC	4-6 weeks	Best for products planning international expansion (CE + BIS in one go)
STQC	Multiple cities	EMC, Safety, IT product testing	6-8 weeks	Government lab, cheapest, but slower turnaround

Rs 4-8L	3-4 Mo	5	2
TOTAL CERTIFICATION BUDGET	PARALLEL TIMELINE	CERTIFICATIONS REQUIRED	PLANNED ATTEMPTS

Certification Strategy

Parallel processing is the key to meeting the 3-4 month timeline:

- Week 1-2:** Pre-compliance EMC testing at ERTL Bangalore (Rs 50K) — catches design issues early
- Week 2-3:** Fix any EMC issues found in pre-compliance
- Week 3:** Submit BIS application + WPC application + Battery test simultaneously
- Week 4-8:** BIS testing (6 weeks) + WPC testing (4 weeks) + Battery testing (4 weeks) run in parallel
- Week 6-8:** SAR testing (2 weeks) — can start while BIS is in progress
- Week 8-12:** Receive all certifications, apply BIS mark to product, update packaging

Pro Tip: Using a pre-certified ESP32-S3 module (which already has FCC and CE certification) significantly simplifies the WPC process. Espressif provides all RF test reports and compliance documents. The WPC approval becomes a paperwork exercise rather than a full testing cycle, saving 2-3 weeks and Rs 50K+ in testing costs.

2.5 Funding Sources & Distribution Channels

Grant & Funding Opportunities

FUNDING SOURCE	AMOUNT	TYPE	ELIGIBILITY	OUR FIT
BIRAC BIG	Up to Rs 50L	Grant (non-dilutive)	Indian startup, health/biotech innovation, prototype stage	Blind assistive AI qualifies as health-tech assistive device. Strong fit for BIG Grant Phase I + II.
DST PRAYAS	Up to Rs 10L	Grant	Hardware prototype, incubator-backed, Indian startup	Prototyping grant for hardware. Need incubator partnership (IIT Hyderabad THub or CIIE IIM-A).
Startup India Seed	Up to Rs 50L	Convertible debt	DPIIT-recognized startup, <2 years old	General-purpose fund. Already DPIIT-registered. Can supplement other grants.
Villgro	Rs 25-50L	Equity (impact fund)	Social impact startup, health/education/accessibility	Blind assistive is textbook social impact. Villgro also provides mentorship + distribution access to NGOs.
NASSCOM Foundation	Rs 10-25L	Grant + mentorship	Tech for social impact, accessibility technology	Accessibility tech focus. NASSCOM has 1500+ member companies — potential enterprise distribution channel.
Ashoka Foundation	Fellowship + network	Non-monetary	Social entrepreneur, scalable innovation	Fellowship opens doors to global assistive tech network. More valuable for distribution than capital.
IIFL Seed Ventures	Rs 1-3 Cr	Equity	Deep tech, hardware, India-first	If we need institutional capital for scaling production to 10K+ units.

Distribution Channels

CHANNEL	PRODUCTS	MARGIN TO US	STRATEGY & NOTES
D2C Website	All products	Full margin	Own Shopify store. Highest margin but limited reach. Use for early adopters, tech enthusiasts, NRI gifting (big market — NRIs buying assistive devices for parents in India). SEO + social media + YouTube reviews drive traffic.
Amazon India	Standard + Pendant	50-55%	FBA (Fulfilled by Amazon) for fast delivery. Target "assistive technology" and "AI gadgets" categories. Amazon's 350M+ monthly visitors provide massive discovery. Budget Rs 50K/month for PPC initially.
Flipkart	Standard + Pendant	50-55%	Important for Tier 2/3 city reach where Flipkart is stronger than Amazon. Flipkart's Supermart delivery network reaches 19,000+ pin codes.
NAB (National Association for the Blind)	Lite + Standard	35-40%	700+ chapters across India. Direct access to blind community. They organize distribution camps, can do bulk procurement. Relationship-driven — need demos at 5-10 chapters to build trust.
ADIP / ALIMCO / GeM	Lite	30-40%	Government procurement via GeM (Government eMarketplace). ALIMCO distributes assistive devices to 100K+ users/year. Once empaneled, orders come automatically from state-level disability departments.
Drone B2B (direct sales)	SDK + Adapter	75-80%	Direct sales to drone manufacturers and operators. 10-20 key accounts targeted. Conference presence at Bharat Drone Mahotsav, AeroIndia. Pilot program with Garuda Aerospace as anchor customer.

Distribution Priority Order:

- Month 1-3:** D2C website + 5 NAB chapters (prove product-market fit with real blind users)
- Month 3-6:** Amazon India + Flipkart (scale consumer reach) + Start ADIP empanelment
- Month 6-9:** ADIP orders start flowing + 2-3 drone company pilots
- Month 9-12:** Enterprise sales (hotels, airports) + Drone SDK general availability

NRI Gifting Opportunity: There's a large untapped market of NRIs (Non-Resident Indians) who want to buy assistive devices for elderly or visually impaired family members in India. A Rs 13,000 device (~\$155) is an easy purchase for someone earning in dollars. D2C website with "Gift to India" option + expedited shipping could be 15-20% of D2C revenue. Target NRI communities on Facebook, LinkedIn, and WhatsApp groups.

2.6 Customer Journey & Growth Engine

Customer Journey: Blind Assistive Device

STEP	STAGE	WHAT HAPPENS	OUR ACTION
1	Awareness	Family member, caregiver, or NGO volunteer learns about OmniPilot through YouTube demo, NAB newsletter, or word-of-mouth from another user.	YouTube demos showing real blind users navigating with device. Partner with accessibility YouTubers. NAB chapter demos. Hindi + English content.
2	Consideration	Caregiver compares with alternatives (OrCam too expensive, WeWalk limited). Reads reviews, checks ADIP eligibility.	Comparison page on website. ADIP eligibility checker tool. WhatsApp support line for queries. Testimonial videos from beta users.
3	Purchase	Orders via D2C/Amazon OR applies through NAB/ADIP for free device.	Multiple purchase paths. EMI options on Amazon. ADIP application assistance via NAB partner. "Gift to India" option for NRIs.
4	Setup	Unboxes device. Setup must work without sight — audio-guided, simple BLE pairing.	Audio-first setup: device speaks instructions on first boot. Braille quick-start card. Video call support for complex cases. OmniPilot app handles pairing + calibration.
5	First Value	User reads a medicine label in Hindi, identifies a Rs 500 note, or gets an obstacle warning within first 10 minutes.	First-run experience optimized for "wow moment" within 60 seconds. Pre-loaded currency model. Demo mode with guided walkthrough. "Try reading this" prompt.
6	Daily Use	Device becomes essential: reading mail, navigating streets, shopping, recognizing faces.	OTA model updates improve accuracy over time. Usage analytics (anonymized) guide feature priorities. Monthly "what's new" audio newsletter via device.
7	Advocacy	User tells other blind friends, NAB chapter promotes at events.	Referral program: "Refer a friend, get 3 months premium free." NAB volunteer incentive program. Social media stories (with consent).

Growth Flywheel

The OmniPilot Growth Engine:

Impact Stories (blind user testimonials) drive **Media Coverage** (newspapers, TV, YouTube) which drives **NGO Partnerships** (NAB, Sightsavers) which drive **ADIP Empanelment** (government orders) which creates **Volume** (lower BOM cost) which enables **Lower Prices** which creates more **Impact Stories**.

This flywheel is self-reinforcing. The first 50 genuine testimonials from blind users are the fuel that starts it spinning. Everything else follows from real impact.

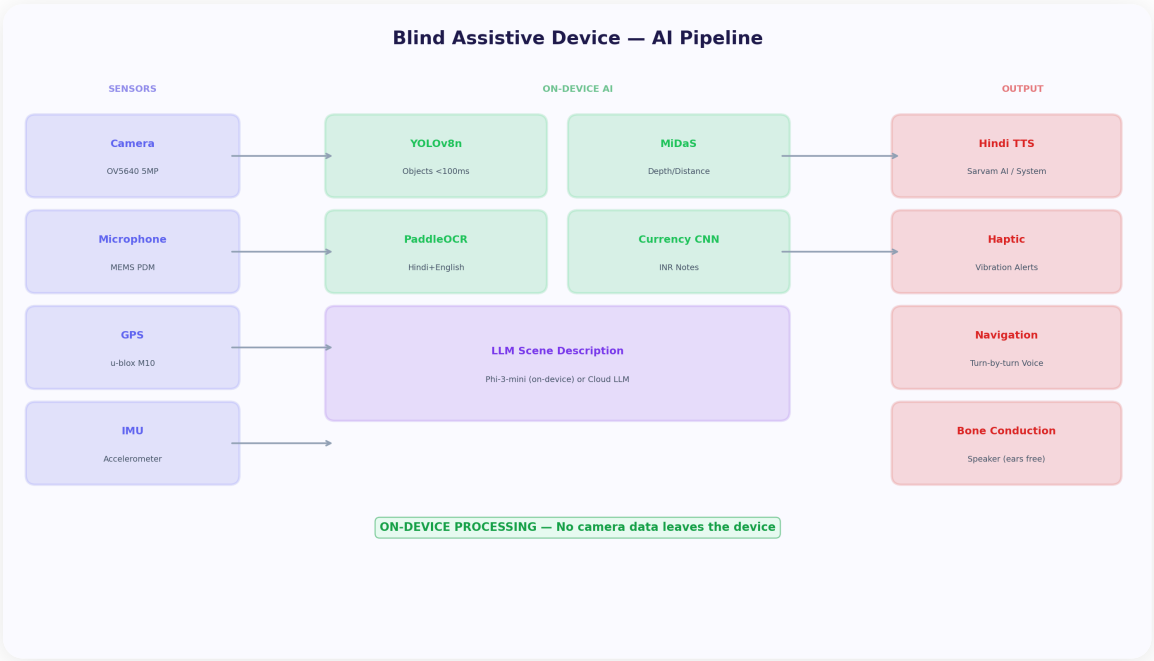
Metrics That Matter (KPIs)

KPI	MONTH 3 TARGET	MONTH 6 TARGET	MONTH 12 TARGET	WHY IT MATTERS
Devices activated	50	300	1,500	Core adoption metric. "Activated" = used for 3+ days.
Daily active usage rate	70%	75%	80%	If people stop using it, the product isn't useful enough.
Object detection accuracy	85%	90%	95%	Safety-critical. Must improve with each OTA update.
NPS (Net Promoter Score)	+40	+50	+60	Word-of-mouth is our primary growth channel.
Cloud AI subscription rate	20%	30%	40%	Recurring revenue from Lite-to-Standard upgrades.

The Single Most Important Metric: Daily active usage rate. If a blind person uses the device every day, we've built something genuinely useful. If they stop after a week, we've built a gadget. Everything else — revenue, partnerships, funding — follows from daily usefulness.

3.1 Blind Assistive Device Architecture

Think of it this way: A pendant with a camera (eyes), a microphone (ears), a speaker (voice), and a vibration motor (touch). The AI running on the phone is the brain. The camera sees the world, the AI understands it, and the speaker describes it in your language. It's like having a sighted friend who constantly tells you what's around you.



Component Architecture

COMPONENT	HARDWARE	FUNCTION	LATENCY	PROCESSING LOCATION
Camera	OV5640 5MP autofocus	Captures scene at 5-15fps depending on mode	<50ms capture	On-device (ESP32-S3 captures frame)
Object Detection	ESP32-S3 + NPU assist	YOLOv8-nano detects people, vehicles, obstacles, animals	<100ms	On-device (quantized INT8 model)
Depth Estimation	MiDaS v3 small	Monocular depth map for obstacle proximity warning	<200ms	Phone (via BLE) or cloud
OCR	PaddleOCR	Reads text in Hindi, English, Tamil, Telugu + 5 more	<300ms	On-device (Lite) or phone (Standard)
Scene Description	LLM (GPT-4o mini / Gemma)	"You are at a crossroad. There's a bus stop on the	1-3s	Cloud only (Standard/Pro tier)

Component	Hardware	Function	Latency	Processing Location
		left with 3 people waiting."		
Currency ID	Custom CNN (3 layers)	Identifies Indian currency notes (Rs 10 to Rs 2000)	<50ms	On-device (embedded model, 200KB)
Face Recognition	FaceNet / ArcFace lite	Recognizes saved faces (family, friends). "Mom is 2 meters ahead."	<500ms	Phone (via BLE) — Pro tier only
Navigation	Phone GPS + Depth	Turn-by-turn walking directions with obstacle avoidance	Continuous	Phone + cloud (Google Maps API)
Hindi TTS	Sarvam AI / eSpeak	Converts AI output to natural Hindi speech	<200ms	Sarvam cloud (Standard) / eSpeak on-device (Lite)
Haptic Feedback	Vibration motor 10mm	Proximity warnings: faster vibration = closer obstacle	<10ms	On-device (direct GPIO)

Data Flow: What Happens When User Presses the Button

1. **User double-taps pendant** → ESP32-S3 wakes camera
2. **Camera captures frame** → 640x480 JPEG, ~30KB
3. **On-device YOLO runs** → Detects objects, bounding boxes (100ms)
4. **If text detected** → PaddleOCR reads Hindi/English text (300ms)
5. **If currency detected** → CNN identifies denomination (50ms)
6. **Lite tier stops here** → eSpeak announces "Person ahead, 2 meters. Sign reads: Dr. Sharma Clinic." (total: 500ms)
7. **Standard tier continues** → Image sent to phone via BLE (200ms) → phone sends to cloud LLM
8. **Cloud LLM generates scene** → "You're standing in front of a medical clinic. The sign says Dr. Sharma in Devanagari. There are two people sitting on a bench to your left." (1-2s)
9. **Sarvam AI converts to Hindi speech** → Natural Hindi voice plays through earpiece

The 500ms Promise: For the Lite tier (offline), the total pipeline from button press to speech output is under 500ms. That's fast enough to feel instant. The Standard tier adds 2-3 seconds for the rich scene description but always plays the basic detection first so the user isn't waiting in silence.

3.2 AI Model Stack & Product Tiers

On-Device AI Models

MODEL	SIZE	INFERENCE	HARDWARE	DETAILS
YOLOv8-nano	3.2MB	<100ms	ESP32-S3	80 COCO classes + 20 India-specific (auto-rickshaw, cow, open manhole, cycle-rickshaw). INT8 quantized. Fine-tuned on 10K Indian street images.
MiDaS v3 small	25MB	<200ms	Phone (ARMv8)	Monocular depth estimation. Converts single image to depth map. Used for proximity warnings: "object 1.5 meters ahead." Runs on companion phone.
PaddleOCR (Hindi)	12MB	<300ms	ESP32-S3 (Lite) or Phone	Devanagari + Latin script OCR. 92% accuracy on Hindi signboards. Quantized for on-device on Lite tier. Full model on phone for Standard.
Currency CNN	200KB	<50ms	ESP32-S3	Custom 3-layer CNN trained on 5K images of Indian currency (Rs 10/20/50/100/200/500/2000). 99.2% accuracy. Tiny model, runs everywhere.
Wake Word (custom)	80KB	Always-on	ESP32-S3	"Hey OmniPilot" or "Bolo OmniPilot" wake word detection using micro speech model. Low power (5mW), always listening.

Cloud AI Models

MODEL	PROVIDER	COST/QUERY	LATENCY	USE CASE
Scene Description	GPT-4o mini / Gemma 3	Rs 0.01-0.03	1-2s	Generates natural language description of camera frame. Prompt: "Describe this scene for a blind person in 2 sentences. Focus on navigation-relevant details."
Hindi TTS	Sarvam AI	Rs 0.005	200ms	Near-native Hindi voice synthesis. Handles code-mixing (Hindi+English) naturally. Voices: male/female, slow/normal speed.
Hindi STT	Sarvam AI / Whisper	Rs 0.01	500ms	Speech-to-text for voice commands. "Yeh kya hai?" (What is this?) triggers camera + scene description.
Translation	IndicTrans2	Rs 0.002	100ms	Translates English OCR text to Hindi/Tamil/Telugu for users who prefer

MODEL	PROVIDER	COST/QUERY	LATENCY	USE CASE
regional language output.				

Product Tiers

FEATURE	LITE (RS 4,999)	STANDARD (RS 12,999)	PRO (RS 24,999)
Connectivity	Fully offline (no phone needed)	BLE to phone + WiFi/4G	BLE + WiFi + 4G SIM slot
Object Detection	YOLOv8-nano (80+20 classes)	YOLOv8-nano + cloud refinement	YOLOv8-small (300+ classes)
OCR	PaddleOCR on-device (Hindi+English)	PaddleOCR + cloud correction (8 langs)	Full multilingual + document OCR
Scene Description	Template-based ("Person ahead, 2m")	LLM natural language (2 sentences)	LLM detailed (5+ sentences, context-aware)
TTS	eSpeak (robotic but functional)	Sarvam AI (natural Hindi)	Sarvam AI (natural, multiple voices)
Navigation	Obstacle detection + haptic only	Google Maps turn-by-turn + obstacles	Full indoor+outdoor navigation
Face Recognition	No	No	Yes (save 50 faces, announce by name)
Battery Life	12 hours (low power)	8 hours (BLE active)	6 hours (full features active)
Camera	2MP OV2640	5MP OV5640 autofocus	5MP + depth sensor (ToF)
ADIP Eligible	Yes	Partial (may exceed limit)	No (above ADIP cap)

Build Path: App First, Hardware Second

Smart Sequencing:

- Month 1:** Build the AI pipeline as an **Android app**. Phone camera = eyes. Phone speaker = voice. This validates the AI stack with zero hardware risk.
- Month 2:** Give app to 20 blind beta users. Collect feedback: what features matter most, what accuracy is needed, what form factor they want.
- Month 3:** Design pendant hardware based on validated app experience. Only build the hardware features that beta users actually use daily.

This "app-first" approach means we never build hardware features nobody wants. The app is the prototype; the pendant is the product.

3.3 BLE Pendant – The Omi Competitor

Technical Specifications

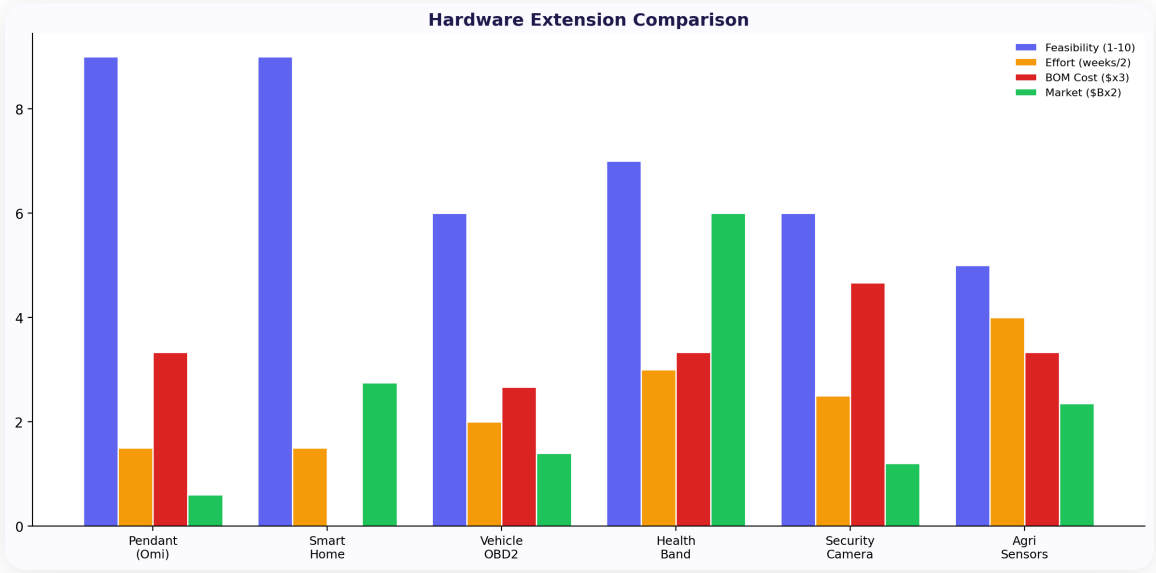
SPEC	VALUE	NOTES
Processor	nRF52840 (ARM Cortex-M4F)	64MHz, 256KB RAM, 1MB Flash. BLE 5.0 native. Ultra-low power (7mA active, 1.5uA sleep). \$3.50 at volume.
Microphone	SPH0645LM4H (MEMS I2S)	Digital I2S output, 65dB SNR, -26dBFS. No ADC needed. Ideal for voice capture in noisy environments.
Battery	200mAh LiPo 3.7V	~18 hours with Opus codec streaming. ~72 hours in standby (wake word only). USB-C charging, 45-min full charge.
Audio Codec	Opus 32kbps (BLE streaming)	Opus provides excellent speech quality at 32kbps. Streams via BLE GATT to phone. Phone does STT + AI processing.
BLE	BLE 5.0, 2Mbps PHY	High-speed PHY for audio streaming. 10m range. Automatic reconnection to paired phone. Custom GATT profile.
LEDs	1x RGB LED	Status indication: blue=connected, green=listening, red=low battery. Subtle, non-distracting.
Button	1x tactile + 1x capacitive touch	Tactile for on/off + recording toggle. Capacitive touch zone for tap gestures (single/double/long press).
Size	32mm diameter x 10mm thick	Approximately the size of a large coin. Lightweight at 12-15 grams with battery.
Charging	USB-C (or pogo pins)	USB-C for standard, pogo pins for premium (magnetic dock). TP4056 charging IC.

BOM Breakdown

COMPONENT	COST (INR)	% OF BOM
nRF52840 module	Rs 292	37%
MEMS Microphone	Rs 100	13%
LiPo Battery 200mAh	Rs 85	11%
PCB + Passives	Rs 125	16%
Enclosure (injection molded)	Rs 100	13%

Component	Cost (INR)	% of BOM
USB-C connector + cable	Rs 45	6%
Misc (LED, button, antenna)	Rs 40	5%
Total BOM	Rs 787	100%

At Rs 787 BOM and Rs 2,499 retail, gross margin is **68.5%**. At Rs 4,999 premium tier, it's **84.3%**. This is the highest-margin product in our portfolio.

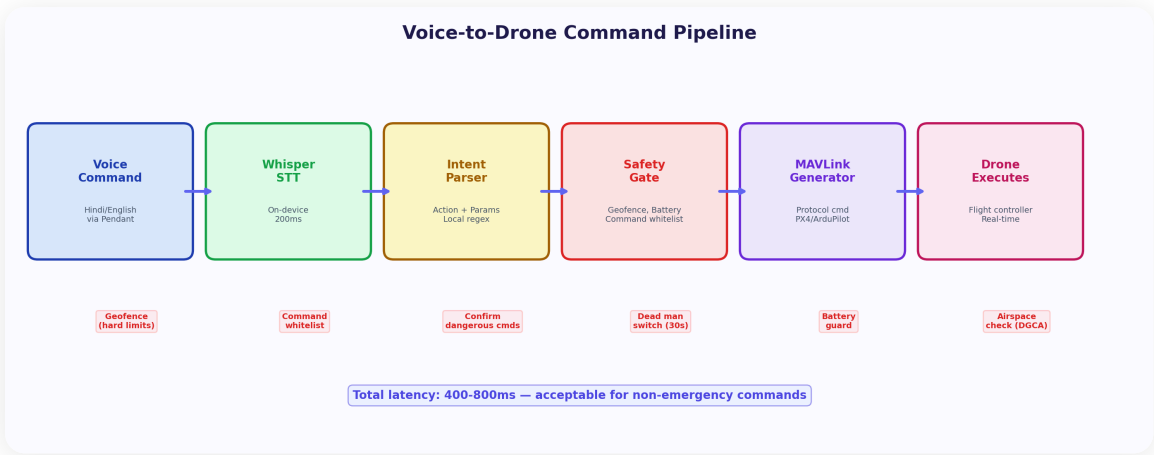


Form Factor Options

Form Factor	Target User	Pros	Cons
Pendant (necklace)	Professionals, students	Always present, fashion-neutral, good mic position near mouth	May snag on clothing, not ideal for sports
Clip-on badge	Office workers, events	Easy to attach/remove, works with any clothing	Falls off occasionally, mic further from mouth
Earpiece	Commuters, fitness	Best mic position, discreet, bone conduction option	Requires separate charging case, higher BOM (+Rs 300)
Wristband	Fitness, elderly	Hard to lose, fitness integration, always accessible	Poor mic position (wrist), limited battery space

Launch Strategy: Start with the **pendant form factor** (lowest tooling cost, simplest enclosure). Add clip-on as a variant with the same PCB. The earpiece form factor requires a separate charging case design — defer to v2 after validating pendant demand. Omi also started as a pendant, confirming this is the right entry form factor.

3.4 Drone Voice Integration Architecture



Command Taxonomy

COMMAND TYPE	EXAMPLE (ENGLISH)	EXAMPLE (HINDI)	PROCESSING
Direct	"Take off"	"Ud jao"	1:1 mapping to MAVLink command. No parameters needed. Instant execution.
Parametric	"Go up 10 meters"	"Das meter upar jao"	Extract numeric parameter + direction. Map to MAVLink altitude change.
Semantic	"Orbit the building"	"Building ke around ghoom"	Requires GPS context + object detection. Plan circular waypoint mission around detected structure.
Contextual	"Go to where you were before"	"Pehle wali jagah pe jao"	Requires session memory. Recall previous waypoint and navigate back.
Mission	"Spray my field starting from the northeast corner"	"Uttar-purv kone se khet mein spray karo"	Complex: parse field boundaries from GPS, plan spray pattern, calculate pesticide flow rate. Multi-step mission.

Safety Architecture (6 Layers)

#	SAFETY LAYER	WHAT IT DOES	OVERRIDE BEHAVIOR
1	Geofence	Prevents drone from flying beyond pre-set GPS boundaries. Auto-return if breached.	Cannot be overridden by voice. Hard limit. Only changeable via app settings with PIN.

#	SAFETY LAYER	WHAT IT DOES	OVERRIDE BEHAVIOR
2	Command Whitelist	Only recognized commands execute. Unknown voice input is rejected with "Command not recognized."	Prevents hallucinated or misheard commands from executing. Whitelist is version-controlled.
3	Confirmation Gate	Dangerous commands require verbal confirmation: "Confirm: fly to 120 meters altitude?"	Timeout = abort. "Cancel" = abort. Only "Confirm" or "Haan" proceeds.
4	Dead Man's Switch	If no voice/BLE contact for 30 seconds, drone hovers in place. 60 seconds = auto-return-to-launch.	Prevents flyaway if operator loses connection. Timer resets on any valid voice input.
5	Battery Guard	Auto-return-to-launch when battery hits 25%. Warning at 35%. No takeoff below 40%.	Cannot be overridden. "I need 5 more minutes" is rejected if battery is critical.
6	Airspace Check	Queries Digital Sky / AirMap API before takeoff. Blocks flight in no-fly zones (airports, military).	Cannot be overridden. Required by DGCA. Displays zone info: "You are 2km from airport. Flight not permitted."

Partner Strategy: SDK Licensing, Not Building Drones

PARTNER	TYPE	THEIR DRONES	OUR INTEGRATION	REVENUE MODEL
Garuda Aerospace	Mfg (India's largest)	Kisan Drone (agri), TSAW delivery	Voice SDK on their GCS + BLE adapter for field ops	Rs 8,999/drone/year + setup fee
IoTechWorld	Agri drone specialist	Agribot, Kissan Drone	Hindi voice commands for farmers ("Spray shuru karo")	Rs 8,999/drone/year
ideaForge	Defense/enterprise	Ninja, Switch, Netra	English voice commands for surveillance operators	Enterprise license Rs 2-5L/year
DJI (via resellers)	Consumer/prosumer	Mavic, Mini, Air series	App-based voice control (no hardware adapter needed)	App subscription Rs 999/year

We Don't Build Drones. We build the intelligence layer. Garuda has manufacturing, distribution, and service networks across India. They don't have AI voice control. We have AI voice control. We don't have drone manufacturing. Partnership > vertical integration. This keeps our CAPEX near zero for the drone vertical.

3.5 Smart Home & Future Extensions

Smart Home Integration (Software-Only)

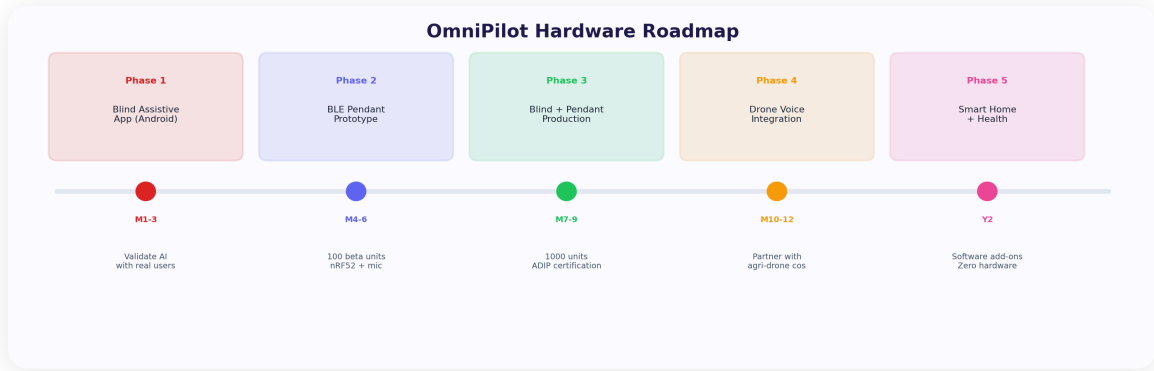
Smart home voice control requires **zero additional hardware**. The OmniPilot app + BLE pendant + existing smart home protocols cover 80% of Indian smart homes. This is a pure software revenue stream.

PROTOCOL	COVERAGE	DEVICES	OMNIPILOT INTEGRATION
Tuya/Smart Life	~80% of Indian smart home devices	Lights, switches, ACs, fans, plugs, cameras from 100+ brands (Wipro, Syska, Orient, Atomberg, etc.)	Tuya Cloud API integration. Control all Tuya devices via OmniPilot voice: "Turn off bedroom light" or "Bedroom ki light band karo."
HomeKit/Matter	~10% (Apple users)	Apple HomePod, Nanoleaf, Eve, select Aqara devices	Matter protocol support via phone. OmniPilot pendant becomes a voice remote for HomeKit. Premium feature.
Google Home	~5%	Nest devices, Chromecast, compatible lights	Google Home API. OmniPilot as alternative voice interface (for users who prefer Hindi voice over Google's).
Alexa	~5%	Echo devices, Ring, compatible devices	Alexa Skills Kit. OmniPilot pendant can trigger Alexa routines via skill integration.

Future Hardware Extensions (Year 2+)

EXTENSION	TIMELINE	BOM EST.	WHAT IT DOES	REVENUE MODEL
Vehicle OBD2	Year 2	Rs 800	Plugs into car OBD2 port. Voice diagnostics: "Car ki health batao" → "Engine oil change needed in 500km. Tyre pressure low: rear left."	Rs 1,999 device + Rs 99/mo subscription
Health Band	Year 2	Rs 1,200	Wrist-worn. Heart rate, SpO2, step count + OmniPilot AI insights. "Aaj 8000 kadam chale, heart rate normal, oxygen 98%."	Rs 3,999 device + Rs 149/mo health insights

EXTENSION	TIMELINE	BOM EST.	WHAT IT DOES	REVENUE MODEL
Security Camera	Year 3	Rs 2,000	Indoor/outdoor camera with OmniPilot scene understanding. "Gate pe koi aaya hai" + face recognition for known visitors.	Rs 4,999 device + Rs 199/mo cloud recording
Baby Monitor	Year 2	Rs 1,500	Camera + sound classification. Distinguishes cry types (hunger, pain, discomfort). Hindi TTS alerts to parents.	Rs 3,499 device
Pet Tracker	Year 3	Rs 600	GPS + accelerometer collar. "Buddy 200 meter door hai, park mein khel raha hai."	Rs 1,999 device + Rs 49/mo GPS tracking



The Platform Play

The Amazon Echo Playbook: Amazon launched Echo as a \$180 speaker in 2014. Today there are 20+ Echo devices across every room in the house — from the Echo Dot (\$30) to Echo Show (\$250) to Ring cameras (\$100) to Blink (\$30). The speaker was the beachhead. The ecosystem is the business.

OmniPilot's beachhead is the blind assistive pendant. The ecosystem is: pendant + drone adapter + smart home + vehicle + health + security. Each device shares the same AI platform, same app, same voice interface. Users who buy one device are pre-sold on the next one because the experience is seamless.

8 PLANNED PRODUCTS BY YEAR 3	1 AI PLATFORM POWERS ALL	Rs 50- 100 Cr YEAR 3 REVENUE POTENTIAL	80% SMART HOME VIA TUYA (NO HW)
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Year 1 Focus: Despite the exciting future extensions, Year 1 is laser-focused on **three products only**: blind assistive device, BLE pendant, and drone SDK. No distractions. Smart home is a software update,

not a product launch. Vehicle, health, and security are Year 2-3 when we have revenue, team, and manufacturing relationships in place.

4.1 90-Day Action Plan

WEEK	FOCUS AREA	KEY DELIVERABLES	SUCCESS CRITERIA
1	Android App MVP	Camera + YOLO object detection + Hindi TTS running on Android phone. Basic UI.	App detects 10+ common objects and speaks their names in Hindi in <2 seconds.
2	OCR + Currency	Add PaddleOCR Hindi text reading + currency detection CNN to app.	Reads Hindi text from signs/labels with 85%+ accuracy. Identifies Rs 100-2000 notes with 95%+ accuracy.
3	Cloud Scene Description	Integrate LLM scene description API. Image upload + natural language response.	2-sentence Hindi descriptions of scenes in <3 seconds. Useful for navigation context.
4	Beta Recruitment	Partner with NAB Bangalore chapter. Recruit 20 blind beta testers.	20 signed-up testers with smartphones. Consent forms. WhatsApp group for feedback.
5-6	Beta Testing Round 1	Deploy app to 20 users. Daily usage monitoring. Weekly feedback calls.	15+ users actively using app daily. Top 5 feature requests documented. Critical bugs fixed in <48 hours.
7	Hardware Design Start	Order ESP32-S3 dev kits + OV5640 cameras. Schematic design for pendant.	Dev kit running YOLO inference at 5fps. Schematic review completed. BOM finalized.
8	Grant Applications	Submit BIRAC BIG application + DST PRAYAS. Register on GeM portal.	Applications submitted with beta test data and user testimonials as evidence of impact.
9	PCB Layout + Prototype	4-layer PCB layout. Order 10 prototype boards from PCB manufacturer.	PCB design reviewed (DRC clean). Prototype boards ordered with 1-week lead time.
10	Pendant Assembly	Assemble 5 prototype pendants. Flash firmware. Basic functional testing.	5 working prototypes: camera captures, BLE streams, speaker plays. Battery lasts 6+ hours.
11	Beta Round 2 (Hardware)	Give pendant prototypes to top 5 beta users. Daily usage for 2 weeks.	Users prefer pendant over phone app for daily tasks. Identify hardware-specific issues (heat, ergonomics, battery).
12	Certification Prep	Pre-compliance EMC testing at ERTL. Submit BIS + WPC applications.	Pre-compliance pass. Applications submitted. Production timeline confirmed with VVDN.

Month-by-Month Summary

MONTH	THEME	KEY OUTPUT	BUDGET NEEDED
Month 1	Build & Validate AI	Android app with YOLO + OCR + TTS + scene description. 20 beta users recruited.	Rs 50K (cloud API costs, dev tools)
Month 2	Test & Iterate	Beta feedback incorporated. Hardware design started. Dev kits ordered.	Rs 1.5L (dev kits, PCB prototypes, beta user support)
Month 3	Hardware Prototype	5 working pendant prototypes. Hardware beta with top users. Certification applications filed.	Rs 3L (prototype assembly, certification fees, EMC testing)

Week 1 is the Most Important Week. If the Android app can detect objects and speak Hindi within 7 days, we have a working proof-of-concept. Everything else — hardware, certification, funding, partnerships — depends on proving that the AI pipeline works for Indian conditions (lighting, languages, objects). Day 7 demo to a blind person is the first real validation moment.

4.2 Budget Allocation & Team Structure

Total Investment: Rs 25-40 Lakhs



Detailed Budget Breakdown

CATEGORY	AMOUNT (INR)	% OF TOTAL	DETAILS
Hardware & BOM			
Dev kits & components	Rs 80,000	2.5%	ESP32-S3 kits, cameras, mics, batteries, breadboards, soldering equipment
PCB prototyping (3 runs)	Rs 1,50,000	4.7%	10 + 50 + 200 unit runs. Includes solder paste stencils and SMT assembly for larger runs.
Injection mold tooling	Rs 2,50,000	7.8%	ABS pendant enclosure mold. Single-cavity mold for 5K-10K unit lifetime. 4-week lead time.
First production run (500 units)	Rs 9,00,000	28.1%	BOM @ Rs 1,800/unit for 500 units. Assembly and testing included.
Certification & Compliance			
BIS certification	Rs 2,00,000	6.3%	Including pre-compliance, formal testing, factory audit waiver, annual fee
WPC + EMC + SAR + Battery	Rs 2,50,000	7.8%	All remaining certifications. Budget for 2 attempts on EMC.
Development			
Cloud AI costs (12 months)	Rs 2,00,000	6.3%	LLM API (GPT-4o mini), Sarvam AI TTS/STT, Google Maps API. ~Rs 3/user/day for 200 active users.
Embedded engineer (contract)	Rs 4,00,000	12.5%	6-month contract for firmware development: ESP32-S3 camera driver, BLE audio streaming, OTA updates.
AI/ML training data	Rs 1,50,000	4.7%	India-specific object detection dataset: 10K labeled images of Indian streets, signs, currency, faces.

Category	Amount (INR)	% of Total	Details
Marketing & Launch			
Website + e-commerce setup	Rs 80,000	2.5%	Shopify D2C store, Amazon/Flipkart listing setup, product photography, video demos.
YouTube + social media	Rs 1,20,000	3.8%	5-10 demo videos with real blind users. Accessibility YouTuber partnerships. Instagram/LinkedIn content.
NAB partnership + events	Rs 50,000	1.6%	Demo devices for 10 NAB chapters. Travel for demo events. Braille marketing materials.
Amazon PPC (3 months)	Rs 1,00,000	3.1%	Sponsored Products ads on Amazon India. Target: "assistive device" and "blind aid" keywords.
Operations			
Packaging & accessories	Rs 75,000	2.3%	Printed boxes, USB-C cables, lanyards, Braille quick-start guides, carry pouches. For 500 units.
Warehousing & logistics	Rs 50,000	1.6%	3PL setup for D2C shipping. Amazon FBA fees for first 200 units.
Contingency (10%)	Rs 1,55,000	4.8%	Unexpected costs: certification re-testing, component price increases, additional prototype runs.
TOTAL	Rs 32,10,000	100%	

Team Structure

Role	Type	Monthly Cost	Responsibility
Founder (You)	Full-time	Sweat equity	Product vision, business development, fundraising, ADIP/government relationships
Embedded Engineer	Contract (6 months)	Rs 65K-80K	ESP32-S3 firmware, camera driver, BLE protocol, power management, OTA updates
AI/ML Engineer	Part-time / intern	Rs 25K-40K	YOLO fine-tuning on Indian data, OCR optimization, model quantization, edge deployment
Business Development	Part-time (Month 3+)	Rs 30K-40K	NAB partnerships, ADIP empanelment, drone company outreach, enterprise sales

Lean Team Philosophy: With 2-3 people + AI tools (Claude for code generation, GitHub Copilot for firmware), we can build the MVP. The key hire is the **embedded engineer** — ESP32-S3 camera + BLE

firmware is specialized work. AI/ML can be handled part-time because we're using pre-trained models (YOLO, PaddleOCR) with fine-tuning, not training from scratch.

4.3 Partnerships & Government Strategy

Key Partnerships

PARTNER	TYPE	WHAT THEY BRING	WHAT WE BRING	ENGAGEMENT PLAN
NAB (National Association for the Blind)	Distribution + Advocacy	700+ chapters across India. Direct access to blind community. Organized distribution camps. ADIP experience. Trust of disabled community.	Technology that helps their members. Volume pricing. Training materials. Revenue share on bulk orders.	Month 1: Demo at NAB Bangalore. Month 2: 5 chapters. Month 3: MOU for bulk procurement. Month 6: Pan-India rollout.
Sarvam AI	Technology (TTS/STT)	India's best Hindi TTS engine. Near-native voice quality. 8+ Indian languages. Low latency API.	Real deployment use case with blind users. Feedback on edge cases (accented speech, noisy environments). Joint PR opportunity.	Month 1: API integration. Negotiate volume pricing (target: Rs 0.003/query at 1M+ queries/month). Co-branded marketing.
VVDN Technologies	Manufacturing	ISO 13485 certified. Camera product experience (IP cameras, dashcams). SMT + box-build. 500-unit MOQ.	Design files, BOM, test specifications. 500-unit initial order with growth to 2000/quarter.	Month 2: Factory visit Gurgaon. Month 3: Prototype run (50 units). Month 4: Production pilot (200 units).
Garuda Aerospace	Drone Partner	India's largest drone company. 2000+ drones deployed. Kisan Drone for agriculture. Government contracts. Pan-India presence.	Voice control SDK that makes their drones easier to operate for farmers. Competitive advantage over other Indian drone companies.	Month 4: Technical demo of voice-controlled Pixhawk. Month 5: Pilot program with 10 Kisan Drones. Month 6: SDK license agreement.
Villgro	Impact Investment	Rs 25-50L impact funding. Mentor network. Distribution access to NGOs. Social impact credibility.	Investable social impact startup. Assistive tech with clear metrics (devices deployed, users helped).	Month 2: Application with beta test data. Month 3: Pitch. Month 4: Due diligence. Month 5: Funding close.

Government Strategy

GOVERNMENT BODY	PROGRAM	VALUE TO US	ACTION REQUIRED
ALIMCO	ADIP Device Empanelment	Free distribution to 100K+ users/year	Submit device specs + pricing + test reports. Present at ALIMCO board meeting. Provide 10 evaluation units.
DGCA	Drone Type Certification	Legal requirement for drone SDK sales	SDK partners (Garuda) handle drone certification. We provide voice control documentation for their safety case.
GeM Portal	Government eMarketplace	Direct government procurement	Register as seller on GeM. List blind assistive device. Respond to ADIP tenders from state disability departments.
BIRAC	BIG Grant (Rs 50L)	Non-dilutive development funding	Submit Phase I application with prototype + beta data. If approved, Phase II for scale-up.
DST	PRAYAS Grant (Rs 10L)	Hardware prototyping funding	Apply through incubator (THub or CIIE). Need incubator endorsement letter.
Startup India	DPIIT Recognition + Seed Fund	Tax benefits + Rs 50L seed	Already DPIIT registered. Apply for Seed Fund through approved incubator.

Government Relationship Playbook:

- 1. Start with NGOs, not government directly.** NAB has existing ADIP relationships. They introduce us to ALIMCO. Going directly to ALIMCO as an unknown startup is a 12-month process. Via NAB, it's 6 months.
- 2. Build evidence first.** Government bodies want proof: beta test data, user testimonials, working prototypes. Our 90-day plan generates all of this before any government engagement.
- 3. Use grants strategically.** BIRAC BIG funds development. ADIP funds distribution. Startup India Seed funds operations. Stack these — they're not mutually exclusive.

The NAB Multiplier: One successful demo at a NAB chapter leads to invitations from 10 more chapters. NAB's annual convention attracts 500+ delegates from across India. A 5-minute demo at the convention could generate 50+ partnership leads, 10+ bulk order inquiries, and media coverage in disability-focused publications. This single event could be our biggest marketing moment of Year 1.

4.4 Success Metrics & Impact Goals

Targets by Timeline

Metric	Month 3	Month 6	Month 12	Year 2
Beta Users	50 blind testers	200 active users	1,000 devices activated	5,000+ users
Devices Sold	0 (beta only)	100 (D2C launch)	1,500 total	10,000+ total
Revenue	Rs 0 (pre-revenue)	Rs 10-15L	Rs 50L+	Rs 3-5 Cr
Gross Margin	N/A	55%	60%	65%
NPS Score	+30 (beta)	+40	+50	+60
Object Detection Accuracy	85%	90%	95%	97%
Daily Active Usage	70%	75%	80%	85%
Cloud Subscription Rate	N/A	20%	35%	45%
NAB Chapters Active	5	20	50	100+
Drone Companies Integrated	0	1 (pilot)	3-5	15-20
Pendant Units Sold	0	50	500	5,000
ADIP Empanelment	Applied	In review	Approved	1000+ units/quarter
Grants Secured	Applied for 2	Rs 10-20L	Rs 30-50L	Equity round
Team Size	2 (founder + contractor)	3	4-5	8-10

Impact Scoreboard

1,000

LIVES CHANGED
(YEAR 1)

8

INDIAN LANGUAGES

Rs 5K

MOST AFFORDABLE AI
DEVICE

FREE

VIA ADIP SCHEME

Year 1 Milestones (Go/No-Go Gates)

GATE	WHEN	CRITERIA	DECISION
G1	Day 30	Android app works: YOLO detects objects, Hindi TTS announces them, 20 beta users recruited	GO: Proceed to hardware. NO-GO: Pivot AI approach (try different model) or pivot target user.
G2	Day 60	Beta users use app daily. Top feature requests align with our hardware plan. Positive NPS (+25+).	GO: Order dev kits, start PCB design. NO-GO: Iterate app for 30 more days before hardware commitment.
G3	Day 90	5 pendant prototypes work. Hardware beta users prefer pendant over phone app. Certification applications filed.	GO: Place 500-unit production order. NO-GO: Redesign hardware, extend timeline by 60 days.
G4	Month 6	100+ devices sold (D2C). At least 1 grant approved. NPS +40+. NAB partnership active.	GO: Scale to 1000+ units, hire BD person. NO-GO: Reduce burn, focus on highest-performing channel only.
G5	Month 12	1500+ total devices. Rs 50L+ revenue. ADIP empanelment approved. Drone pilot with 1 company.	GO: Raise seed round. Expand to 3 products. NO-GO: Stay focused on blind assistive only, defer other verticals.

This is About Impact.

Every number on this page represents a real person. 1,000 devices in Year 1 means 1,000 people who can now read their medicine labels, identify currency notes, navigate streets safely, and describe the world around them — in their own language.

The business case is strong: 60%+ margins, multiple revenue streams, government funding. But the real measure of success is simpler:

1 device = 1 person who can see through AI.

If we build something genuinely useful for 50 people in the first 3 months, the rest will follow. Impact is the engine. Revenue is the exhaust.

Technical Appendix: BOM & Manufacturing Details

Detailed Component Sourcing

Component	Supplier	MOQ	Lead Time	Unit Price	Alternatives
ESP32-S3-WROOM-1	Mouser India / LCSC	100	2-3 weeks	\$3.50	ESP32-S3-MINI-1 (\$2.80, smaller antenna). Allwinner V851s (\$4, better NPU but less community support).
OV5640 Camera Module	Aliexpress / Taobao agent	50	2-4 weeks	\$4.00	OV2640 (\$2.50, 2MP, for Lite tier). GC2145 (\$2.80, 2MP, wider availability).
SPH0645LM4H MEMS Mic	Digi-Key India	100	1-2 weeks	\$1.20	INMP441 (\$0.90, I2S, lower SNR but adequate). ICS-43434 (\$1.50, better SNR).
nRF52840 Module	Mouser India	100	2-3 weeks	\$3.50	nRF52833 (\$2.80, BLE 5.1 but less RAM). ESP32-C3 (\$1.50, WiFi+BLE but higher power).
LiPo 1200mAh (pendant)	Local supplier (Bangalore)	200	1 week	\$2.00	800mAh (\$1.50, lighter, 6hr battery). 1500mAh (\$2.50, heavier, 10hr battery).
LiPo 200mAh (BLE pendant)	Local supplier	500	1 week	\$1.00	150mAh (\$0.80, 12hr). 300mAh (\$1.30, 24hr but thicker form factor).
4-layer PCB (pendant)	PCBWay / local (Bangalore)	10	5-7 days	\$2.50	2-layer (\$1.50) possible if EMC is managed carefully. 4-layer recommended for first production.

Manufacturing Cost at Scale

Volume	BOM/unit	Assembly/unit	Total/unit	Savings vs 500-unit
100 units (prototype)	Rs 2,200	Rs 600	Rs 2,800	Baseline (expensive)
500 units	Rs 1,836	Rs 375	Rs 2,211	Reference price
1,000 units	Rs 1,650	Rs 300	Rs 1,950	-12% (volume pricing kicks in)

VOLUME	BOM/UNIT	ASSEMBLY/UNIT	TOTAL/UNIT	SAVINGS VS 500-UNIT
5,000 units	Rs 1,450	Rs 220	Rs 1,670	-24% (component reel pricing)
10,000 units	Rs 1,300	Rs 180	Rs 1,480	-33% (dedicated line pricing)

Quality Assurance Protocol

TEST	METHOD	PASS CRITERIA	SAMPLE RATE
Functional Test	Automated test jig: camera, BLE, speaker, mic, vibration, battery	All peripherals respond within tolerance	100% (every unit)
Camera Quality	Capture test pattern at 1m distance	Focus sharpness > threshold, color accuracy within 5%	100%
BLE Range	Connect at 10m distance through 1 wall	Stable connection, <5% packet loss	10% (spot check)
Battery Cycle	Full charge + discharge cycle, measure capacity	Capacity > 90% of rated (1200mAh → >1080mAh)	5% (destructive)
Drop Test	1.5m drop onto concrete, 3 orientations	Device functional after drop. No battery displacement.	3 units per production batch
Audio Quality	Play test tone, measure dB at 30cm	>70dB SPL, no distortion at max volume	100%
Firmware Verification	OTA update test + version check	Correct firmware version, OTA completes without error	100%

Scale Economics: At 10,000 units, our landed cost drops to Rs 1,480/unit. At the D2C price of Rs 12,999, that's **88.6% gross margin**. Even at ADIP pricing of Rs 4,500, margin improves to **67.1%**. The business gets dramatically better at scale. The first 500 units are about proving the product; units 501-10,000 are about building the business.

OmniPilot Hardware

Voice assistant. Blind assistive AI. Drone control.

Made in India. For India.

Affordable AI hardware for the underserved.
Rs 4,999. Eight Indian languages. ADIP eligible.



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